



Data Infrastructure for Machine Learning

(v1.00)

Week 9: October 28

Jimmy Lin
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University of Waterloo

These slides are available at <http://lintool.github.io/bigdata-2025f/>

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Key Questions

How do data products with operational requirements complicate lakehouse architectures?

Why does retrieval remain important in the era of LLMs?

How is retrieval formulated as the problem of computing vector similarity?

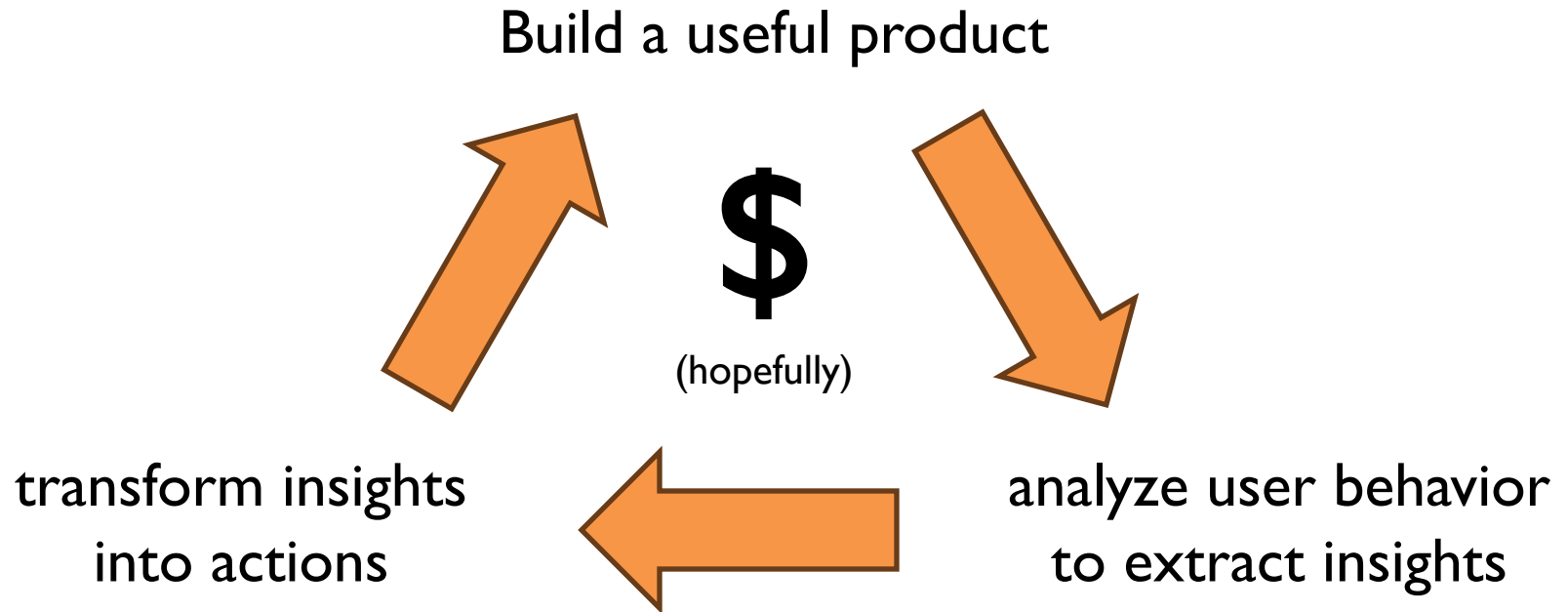
What's the intuition behind sparse and dense representations?

Recap: What are we doing and why?

Context...

The Data Flywheel

(a virtuous cycle)



Google. Facebook. Twitter. Amazon. Uber.

Context...

What's this course about?

The *infrastructure* that supports the data flywheel.

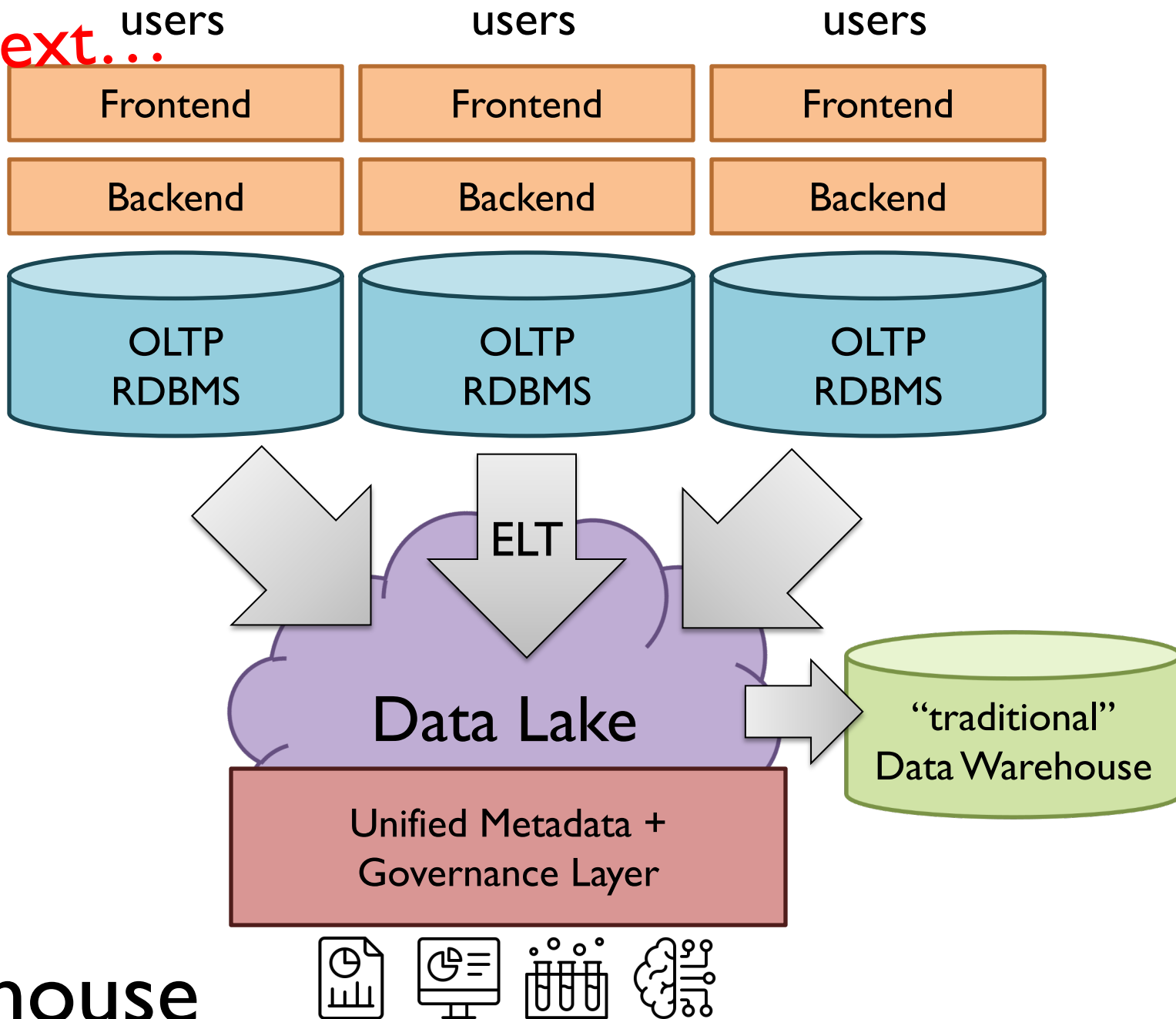
data platforms + data engineering

Context...

What problems do data platforms solve?

Ingesting, storing, manipulating, maintaining, serving...
the data that supports the data flywheel.

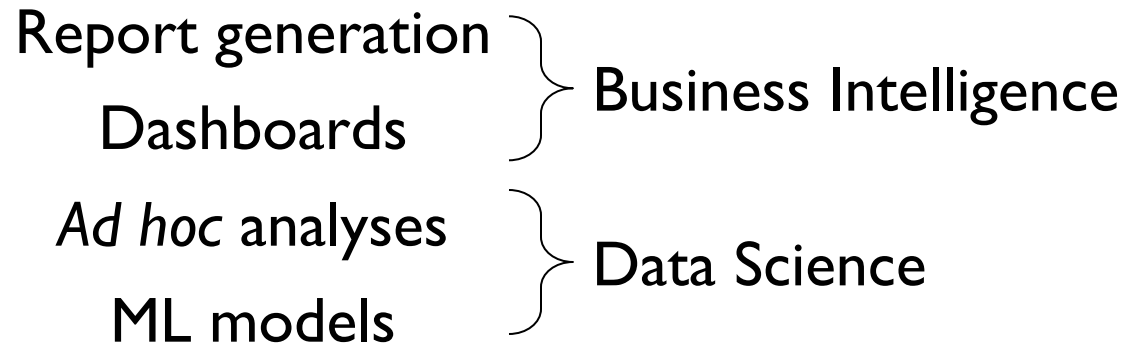
Context...



Context...

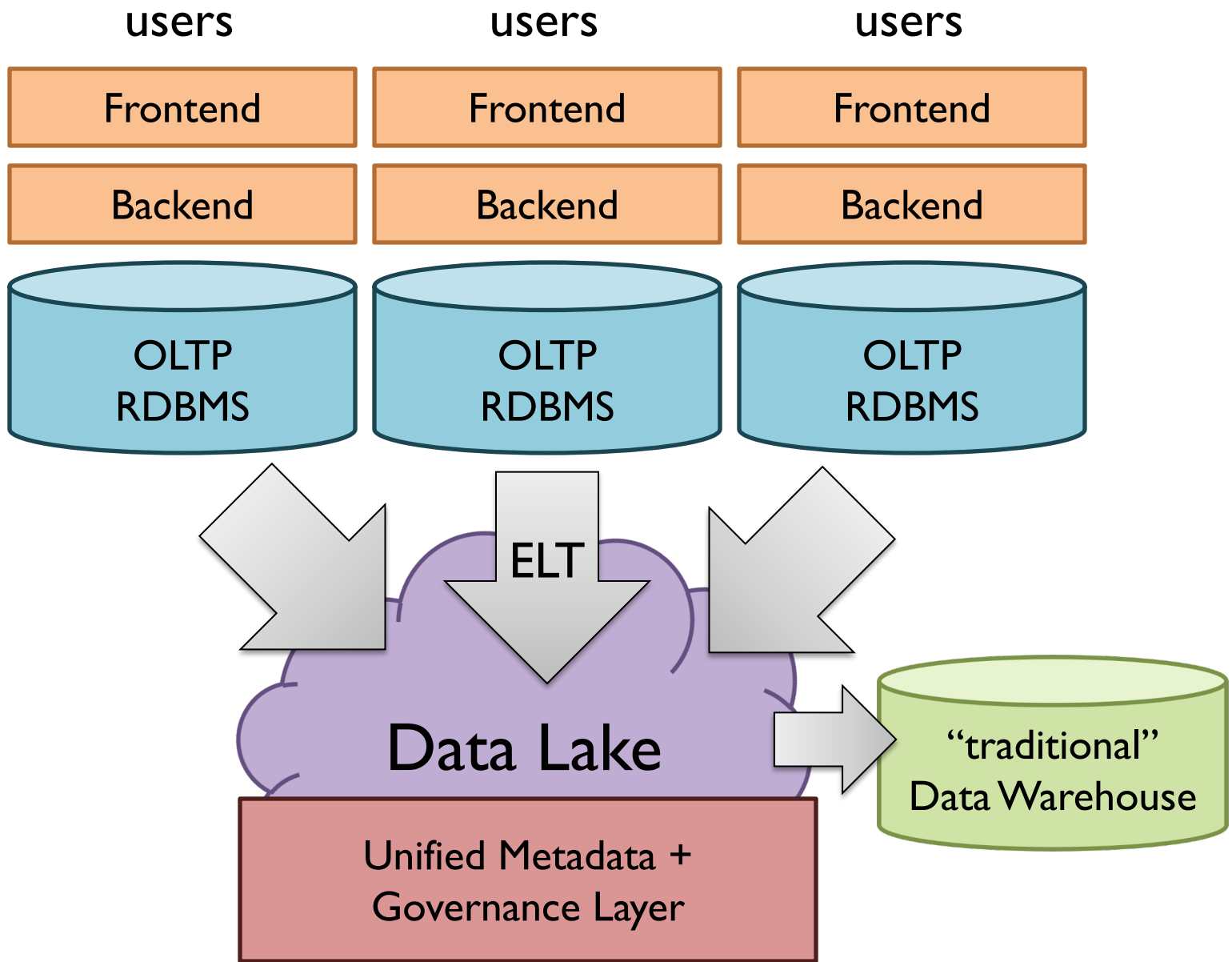
Transform Insights into Actions

What does that really mean?



Syllabus

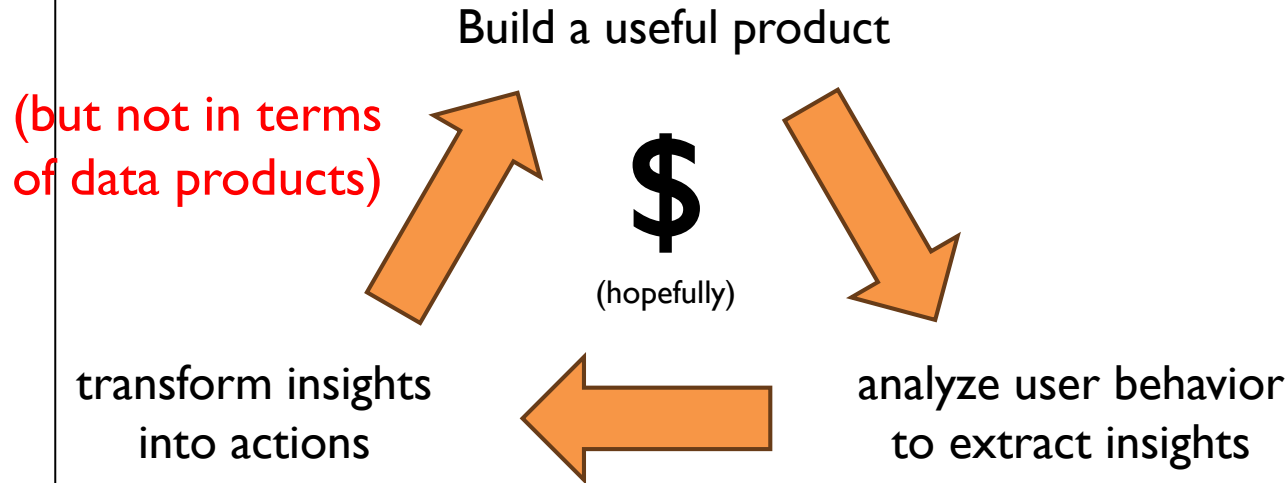
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2	Data Warehouses, Data Lakes, and Lakehouses
3	Batch Processing I
4	Batch Processing II
5	Rubber, Meet Road
6	Data Infrastructure for Machine Learning
7	Reading Week: No Classes!
8	Midterm Exam
9	Text Processing I
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11	Clustering
12	Graph Processing
13	Stream Processing
14	LLMs
	Final Exam



Lakehouse

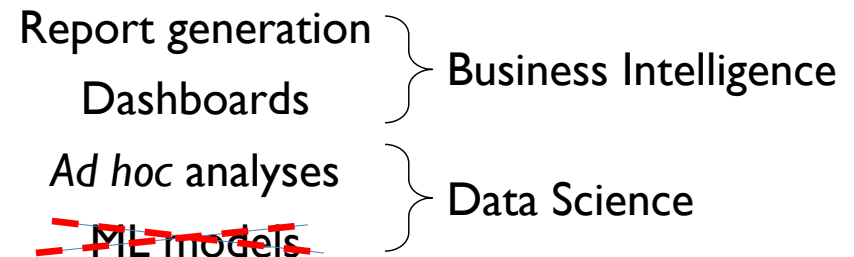
The Data Flywheel

(a virtuous cycle)



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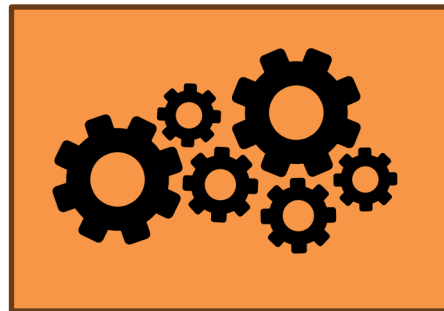
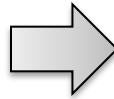


Syllabus

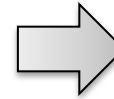
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Instance

amazing spot for good
food & a fun time 🍕🍝
they offer a super unique
dine-in experience with
their interactive tables!
also love that they have
innovative weekly feature
dishes 😊



Model

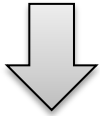


Prediction



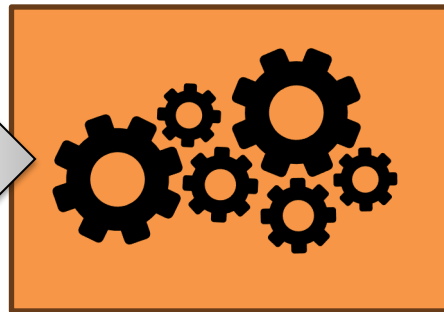
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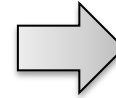
$$\mathbf{x}_i = [x_1, x_2, x_3, \dots, x_d]$$

Feature Vector



Model

$$f : X \rightarrow Y$$



Prediction

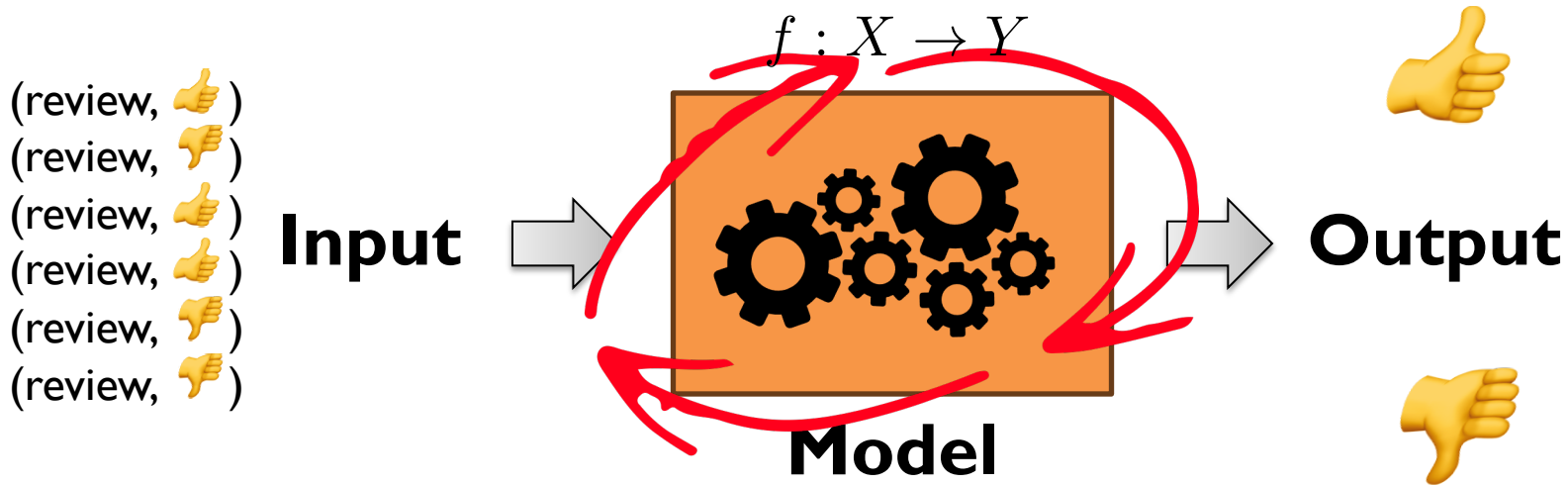


$$y \in \{0, 1\}$$

Components of an ML solution

(data, features, model, optimization)

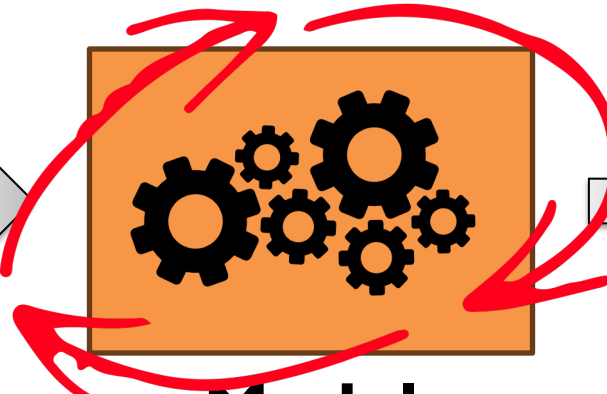
Model *learns* from the data



Machine learning algorithm
adjusts the model *parameters*

(review, 👍)
(review, 👎)
(review, 👍)
(review, 👍)
(review, 👎)
(review, 👎)

Input



Trained **Model**



Output

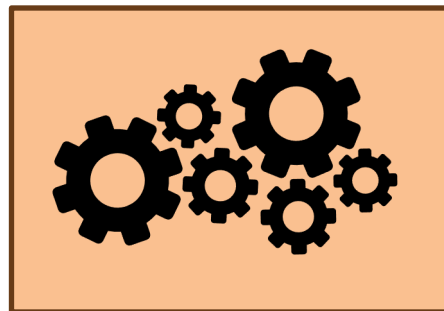
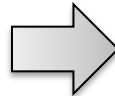


Deployment

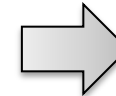


Inference / Prediction

A group of us stopped by
yesterday afternoon to
enjoy an outdoor lunch.
The food was da bomb.



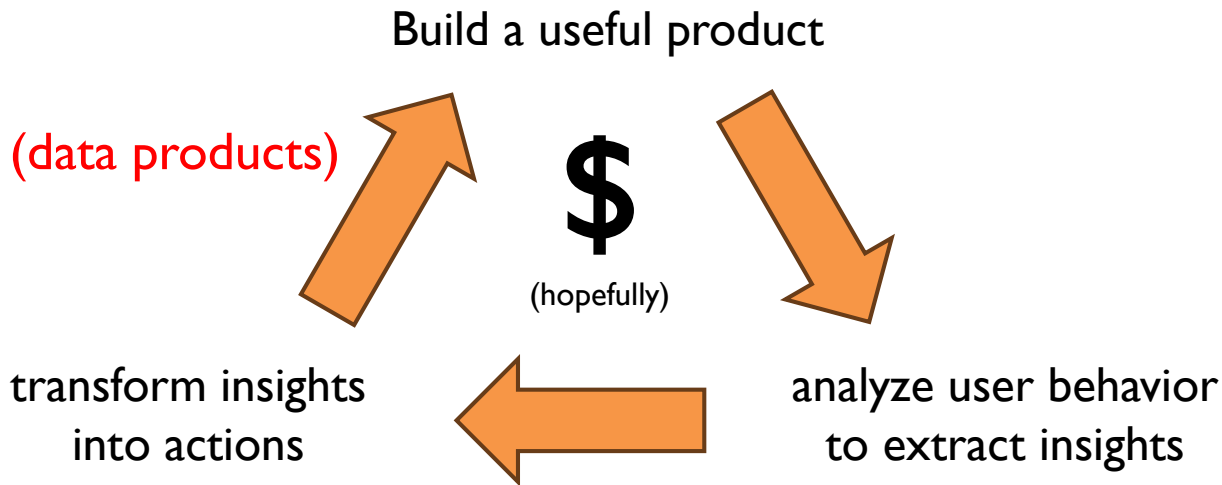
Trained **Model**



Where might you actually deploy this?

The Data Flywheel

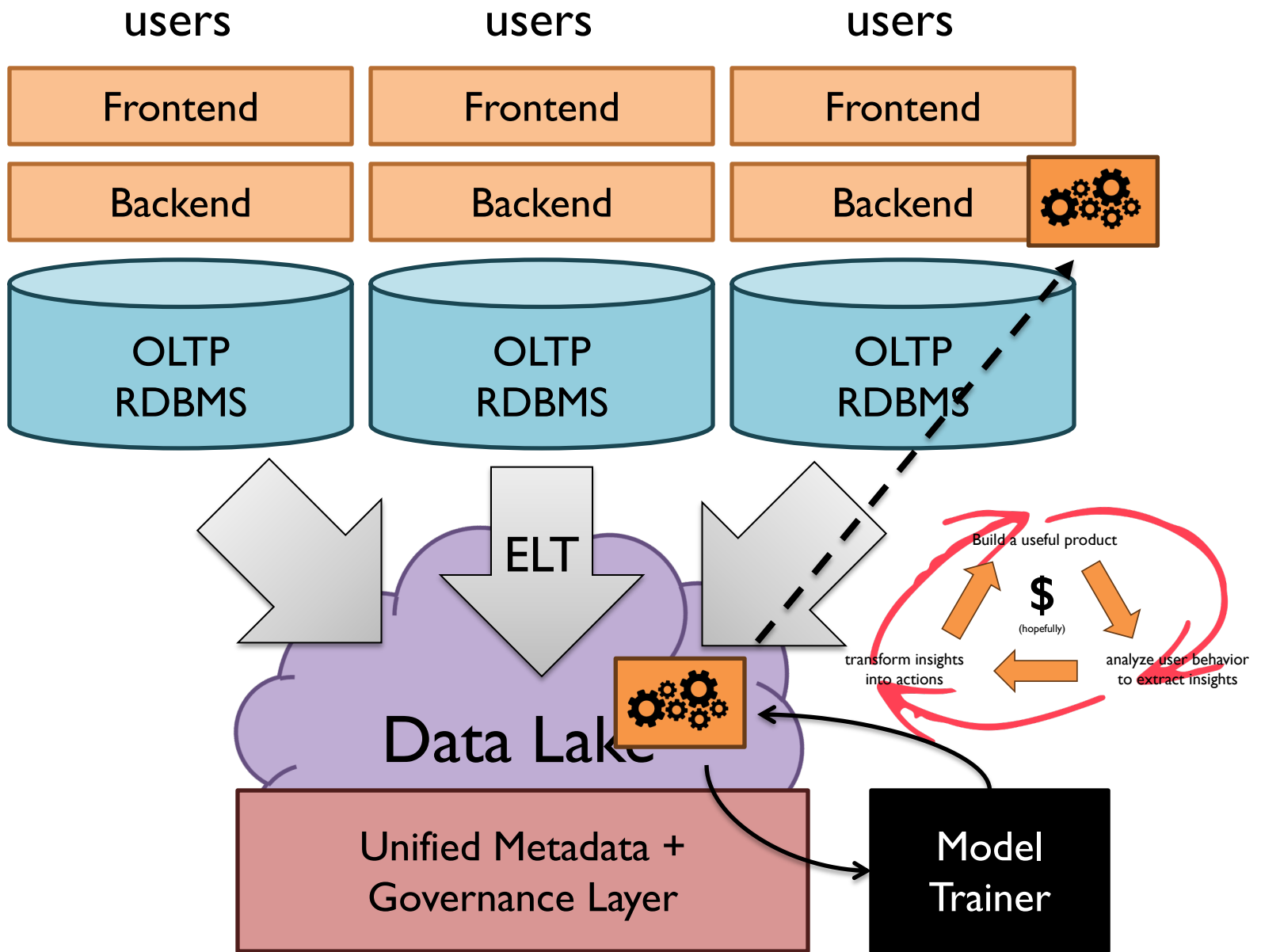
(a virtuous cycle)



Transform Insights into Actions

What does that really mean?





Lakehouse



Syllabus

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Remainder of the Semester

“Weird stuff” that doesn’t fit *traditional* lakehouses

Data products with operational requirements

Both!

Recap: What are we doing and why?
Also: Where are we going and why?

What's the problem we're trying to solve?

How to connect users with relevant information

search (information retrieval) ...
... but also question answering, summarization, etc.
“information access”

... on text, images, videos, etc.
... for “everyday” searchers, domain experts, etc.

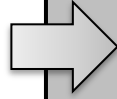
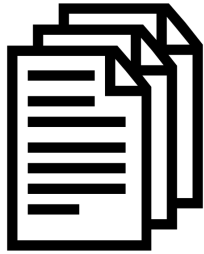


What's the problem we're trying to solve?

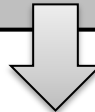
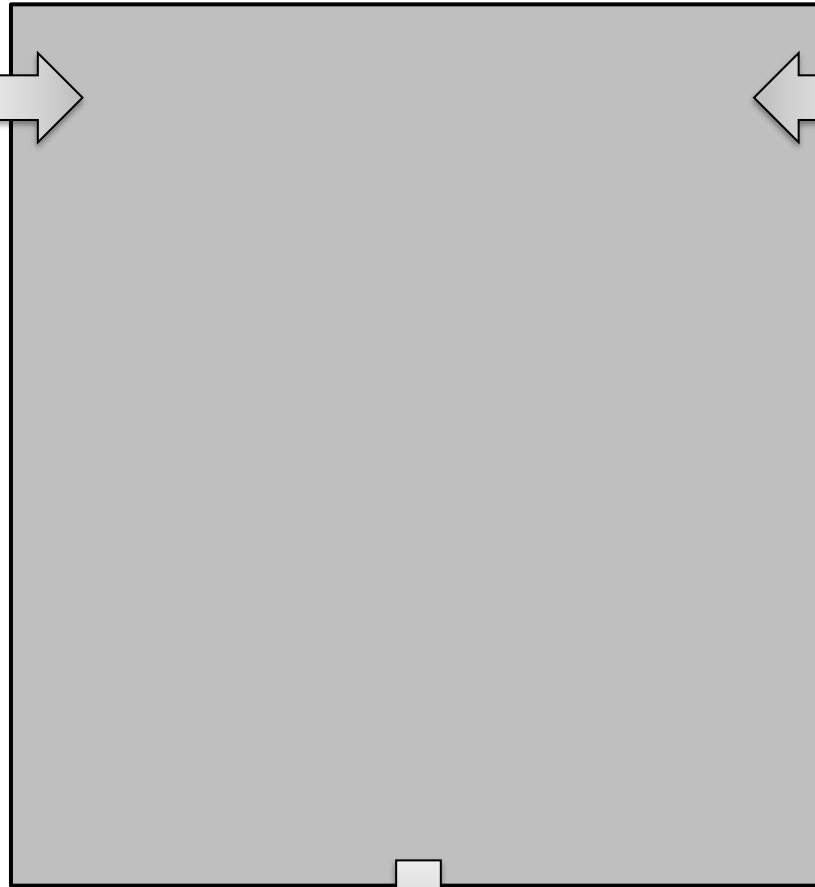
The “core” (text) retrieval problem

Given an information need expressed as a query q , the text retrieval task is to return a ranked list of k texts $\{d_1, d_2 \dots d_k\}$ from an arbitrarily large but finite collection of texts $C = \{d_i\}$ that maximizes a metric of interest, for example, precision, nDCG, AP, etc.

“Documents”



Query



Results

BORING

makeameme.org

That is so last millennium!



Search the web using Google!

10 results



Google Search

I'm feeling lucky

Index contains ~25 million pages (soon to be much bigger)

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10 results

Showing results **1-10** of approximately **234,000** for **google**. Search took **0.06** seconds.

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[GoogleScout](#)

[Google Help](#)

...Basic Search To enter a query into **Google**, just type in a few descriptive...

...descriptive keywords and click on the **Google** Search button for your list...

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[google.stanford.edu/](#)

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[Constructing and refining searches in Google: Detailed Searching Instructions](#)

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...University of California, Berkeley **Google** is a fairly new Web searching...

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A MESSAGE FROM OUR CEO

Questions, shrugs and what comes next: A quarter century of change

Sep 05, 2023 · 9 min read



Sundar Pichai

CEO of Google and Alphabet

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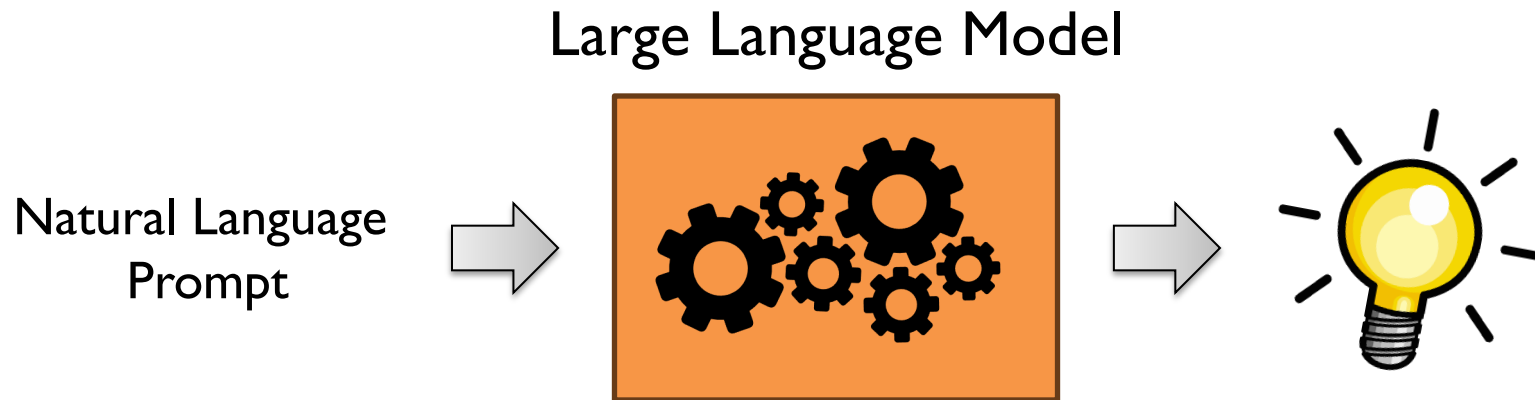
BORING

makeameme.org

But you know what's *not* boring?



What's *not* boring?





Microsoft Bing SEARCH CHAT

I am planning a trip for our anniversary in September. What are some places we can go that are within a 3 hour flight from London Heathrow?

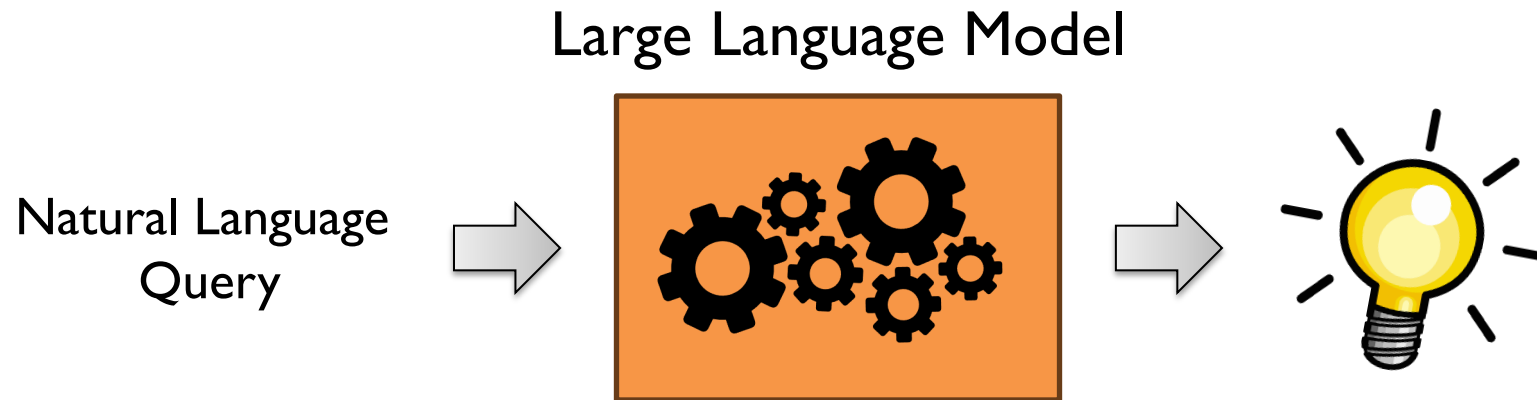
Congratulations on your anniversary! 🎉 There are many places you can go that are within a 3 hour flight from London Heathrow. Here are some suggestions based on your preferences and the best destinations in Europe in September ⁴ ⁵ ⁶:

- If you like **beaches** and **sunshine**, you can fly to **Malaga** in Spain ¹, which has a pleasant climate, a lively culture, and a beautiful coastline. You can enjoy the sandy beaches, the historic monuments, and the delicious tapas. Malaga is also a great base to explore other nearby attractions, such as Granada, Seville, and Cordoba.
- If you like **mountains** and **lakes**, you can fly to **Annecy** in France ⁴, which is a charming alpine town near the Swiss border. Annecy is known as the "Venice of France" because of its canals and bridges. You can admire the stunning scenery, visit the medieval castle, and try some local cheese and wine. Annecy is also close to Geneva, Chamonix, and Lyon, if you want to see more of the region.
- If you like **art** and **history**, you can fly to **Florence** in Italy ⁶, which is the birthplace of the Renaissance and a UNESCO World Heritage Site. Florence is a treasure trove of artistic and architectural masterpieces, such as the Duomo, the Uffizi Gallery, and the Ponte Vecchio. You can also explore the Tuscan countryside, taste the famous gelato, and shop for leather goods.

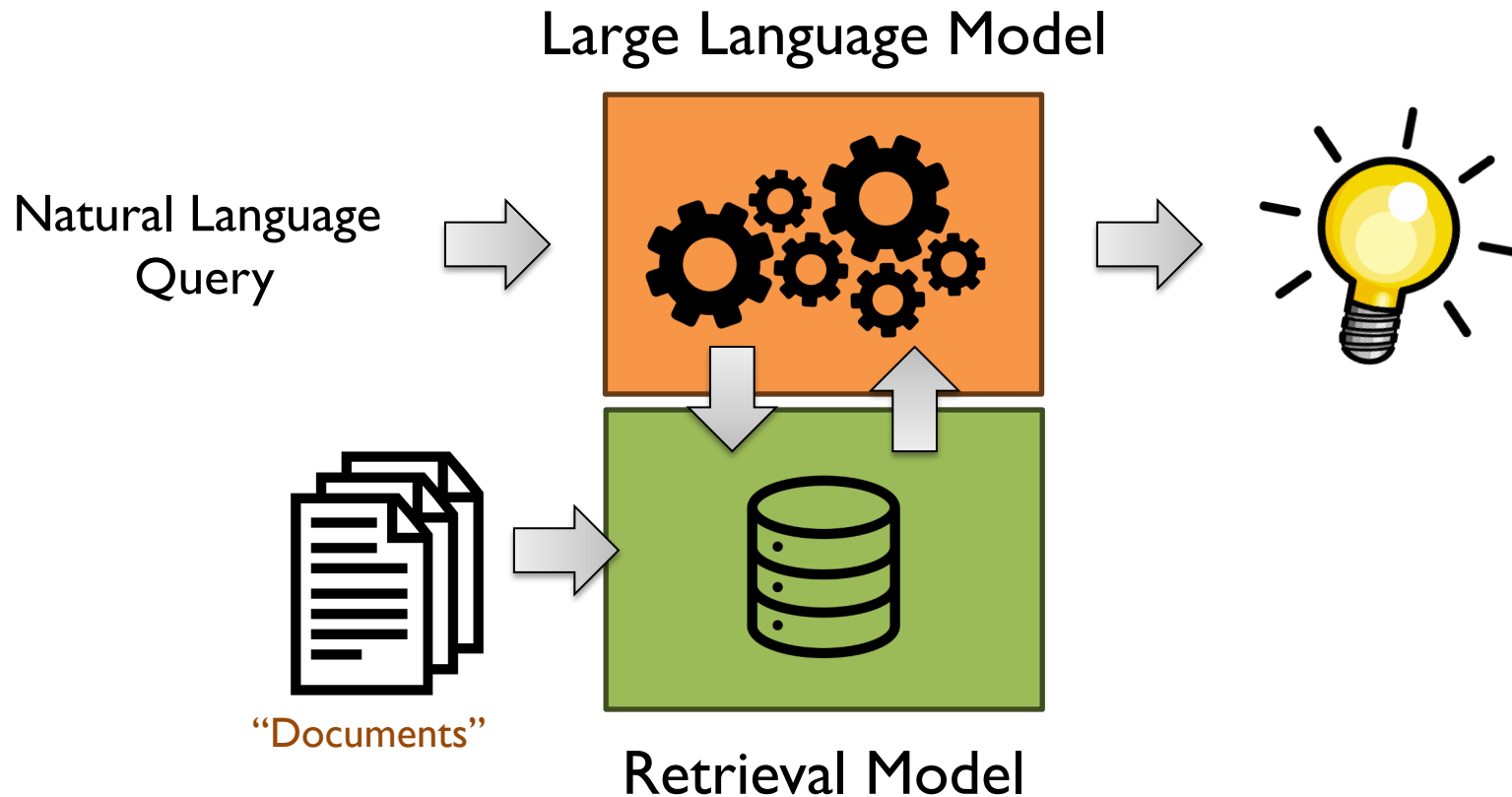
Ask me anything...

Feedback

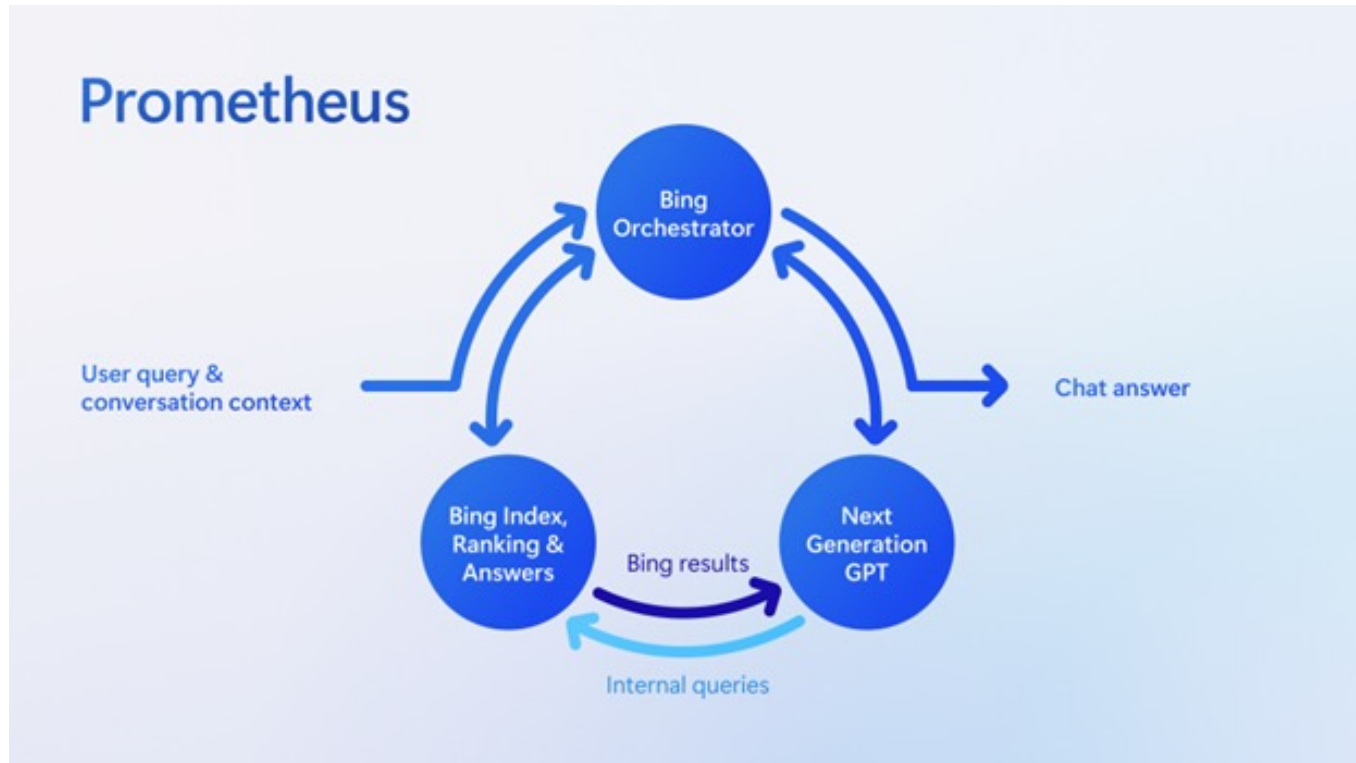
How does Sydney work?



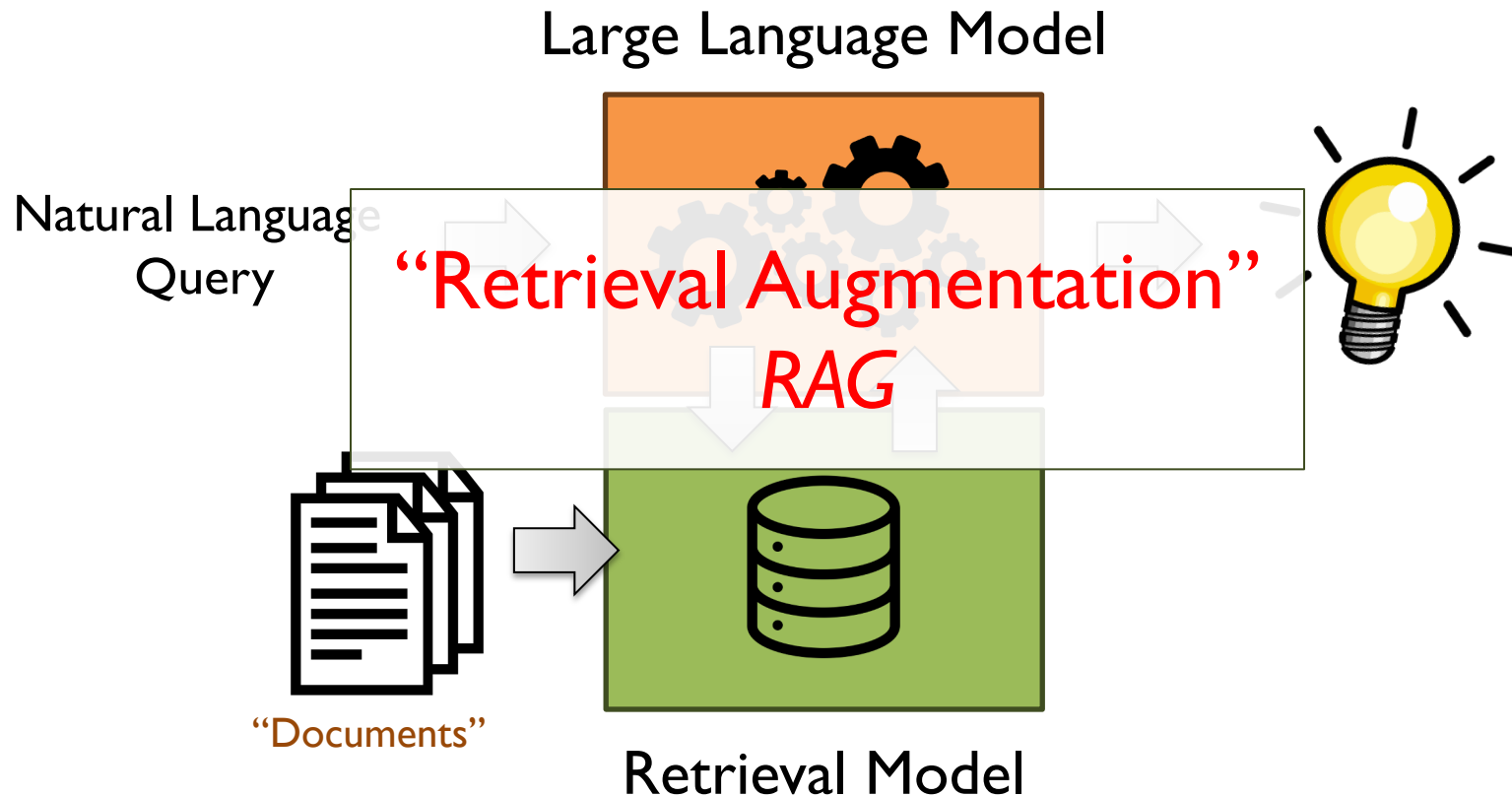
How does Sydney work?



How does Sydney work?



How does Sydney work?



Why Retrieval Augmentation?

(combat) Hallucinations

(incorporate) Up-to-date information

(exploit) Private data

Why Retrieval Augmentation?

Generate a short biography of Jimmy Lin.

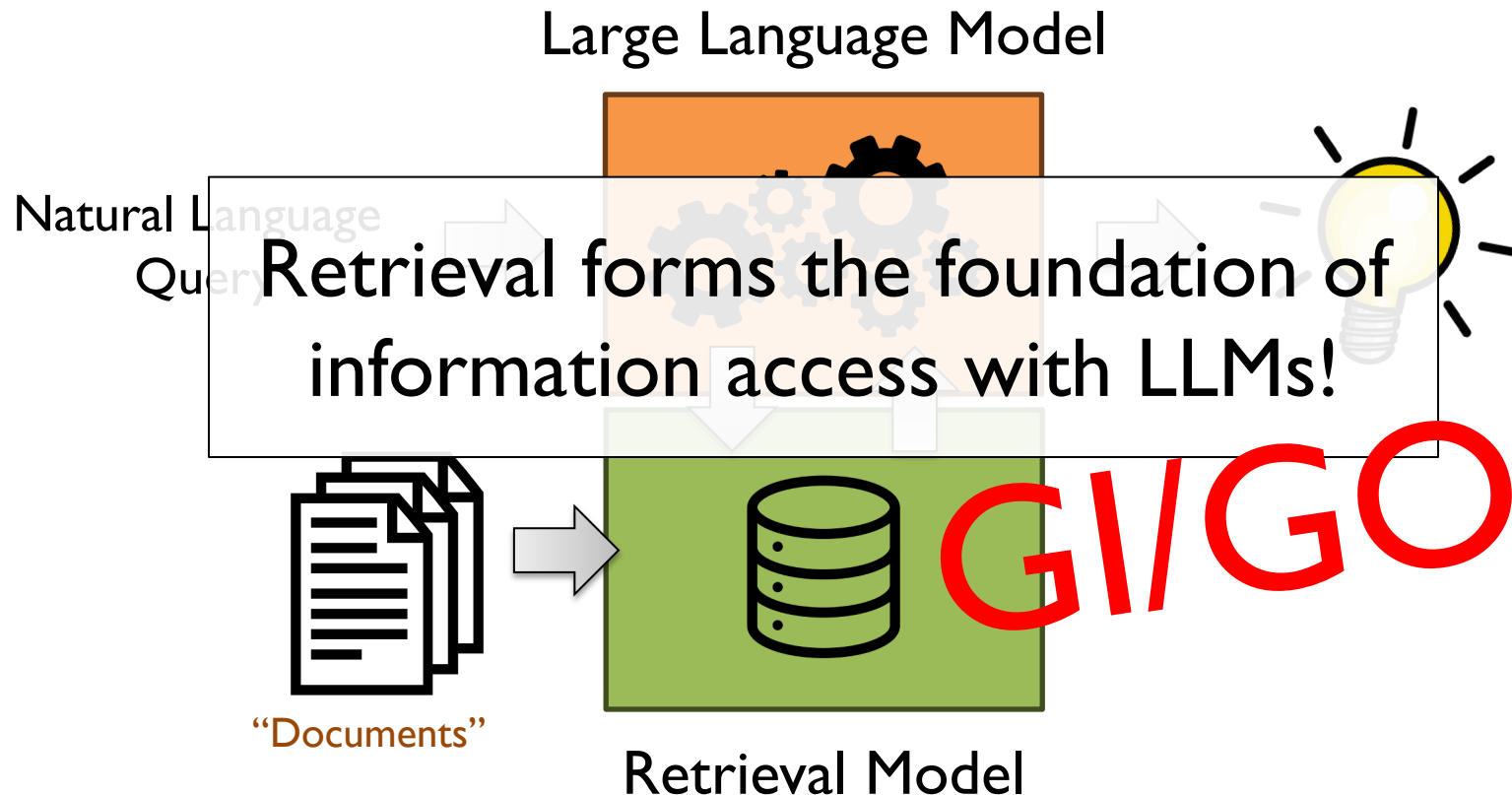
Given the following facts, generate a short biography of Jimmy Lin.

- Jimmy Lin is a professor at the University of Waterloo.
- Lin holds the David R. Cheriton Chair in the David R. Cheriton School of Computer Science.
- Jimmy Lin received his Ph.D. from MIT in 2004
- ...

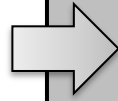
- He recently joined Yupp as Chief Scientist...

- His cell phone number is (519) 721-xxxx...

Why Retrieval Augmentation?



“Documents”



Query



The “core” (text) retrieval problem

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Results

GI/GO

“Documents”



Query



How?



Results

A Vector Space Model for Automatic Indexing

G. Salton, A. Wong
and C. S. Yang
Cornell University

In a document retrieval, or other pattern matching environment where stored entities (documents) are compared with each other or with incoming patterns (search requests), it appears that the best indexing (property) space is one where each entity lies as far away from the others as possible; in these circumstances the value of an indexing system may be expressible as a function of the density of the object space; in particular, retrieval performance may correlate inversely with space density. An approach based on space density computations is used to choose an optimum indexing vocabulary for a collection of documents. Typical evaluation results are shown, demonstrating the usefulness of the model.

Key Words and Phrases: automatic information retrieval, automatic indexing, content analysis, document space

CR Categories: 3.71, 3.73, 3.74, 3.75

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This study was supported in part by the National Science Foundation under grant GN 43505. Authors' addresses: G. Salton and A. Wong, Department of Computer Science, Cornell University, Ithaca, NY 14850; C. S. Yang, Department of Computer Science, The University of Iowa, Iowa City, IA, 52240.

¹ Although we speak of documents and index terms, the present development applies to any set of entities identified by weighted property vectors.

² Retrieval performance is often measured by parameters such as *recall* and *precision*, reflecting the ratio of relevant items actually retrieved and of retrieved items actually relevant. The question concerning optimum space configurations may then be more conventionally expressed in terms of the relationship between document indexing, on the one hand, and retrieval performance, on the other.

1. Document Space Configurations

Consider a document space consisting of documents D_i , each identified by one or more index terms T_j ; the terms may be weighted according to their importance, or unweighted with weights restricted to 0 and 1.¹ A typical three-dimensional index space is shown in Figure 1, where each item is identified by up to three distinct terms. The three-dimensional example may be extended to t dimensions when t different index terms are present. In that case, each document D_i is represented by a t -dimensional vector

$$D_i = (d_{i1}, d_{i2}, \dots, d_{it}),$$

d_{ij} representing the weight of the j th term.

Given the index vectors for two documents, it is possible to compute a similarity coefficient between them, $s(D_i, D_j)$, which reflects the degree of similarity in the corresponding terms and term weights. Such a similarity measure might be the inner product of the two vectors, or alternatively an inverse function of the angle between

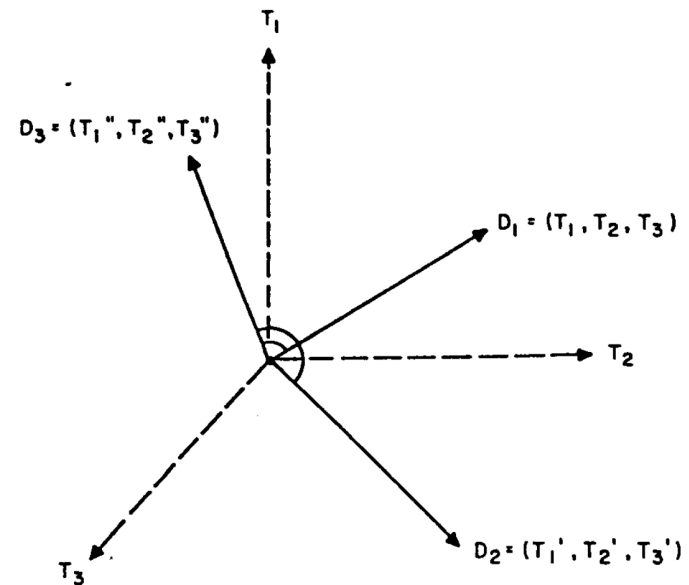
term assignment will be zero, p

Instead of vector original tem, the relaserved by no considering thevelope of the that case, each point whose p corresponding of the space. are then repr together in the tween two do correlated with ing vectors.

Since the function of th are assigned one may ask configuration optimum retr

If nothing under consid ideal docum jointly releva together, thus jointly in res, trariwise, do

Fig. 1. Vector representation of document space.



A Vector Space Model for Automatic Indexing

1. Document Space Configurations

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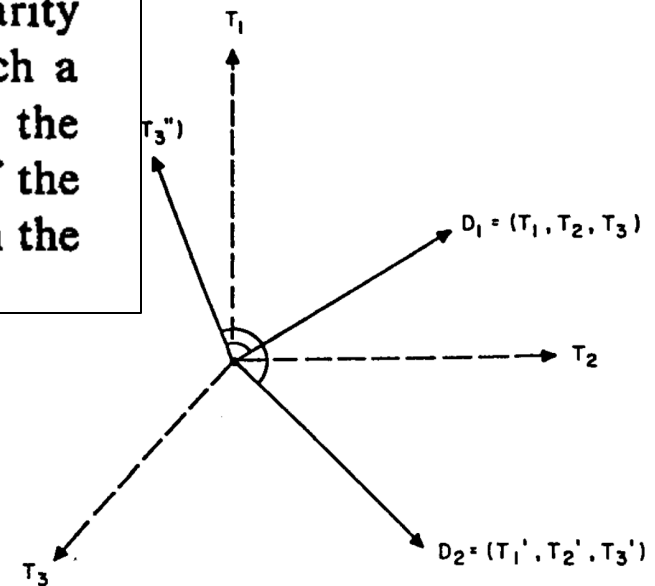
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term. For documents, it is efficient between degree of similarity weights. Such a product of the function of the

presentation of document space.



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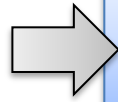
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If nothing under consideration ideal documents jointly relevant together, thus jointly in respect to each other, do

“Documents”

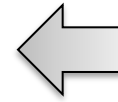


Term Weighting



[...]

Query



The Manhattan Project and its atomic bomb helped bring an end to World War II. Its legacy of peaceful uses of atomic energy continues to have an impact on history and science.

```
{'atom': 4.0140, 'bomb': 4.0704, 'bring': 2.7239, 'continu':  
2.4331, 'end': 2.1559, 'energi': 2.5045, 'have': 1.0742, 'help':  
1.8157, 'histori': 2.4213, 'ii': 3.0998, 'impact': 3.0304, 'it':  
2.0473, 'legaci': 4.1335, 'manhattan': 4.1345, 'peac': 3.5205,  
'project': 2.6442, 'scienc': 2.8700, 'us': 0.9967, 'war':  
2.6454, 'world': 1.9974}
```

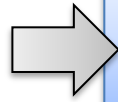
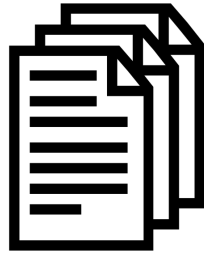


Results

“bag of words”
“lexical”
sparse vector

“Documents”

$$\text{BM25}(q, d) = \sum_{t \in q \cap d} \log \frac{N - \text{df}(t) + 0.5}{\text{df}(t) + 0.5} \cdot \frac{\text{tf}(t, d) \cdot (k_1 + 1)}{\text{tf}(t, d) + k_1 \cdot (1 - b + b \cdot \frac{l_d}{L})}$$

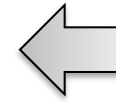


Term Weighting



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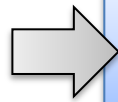
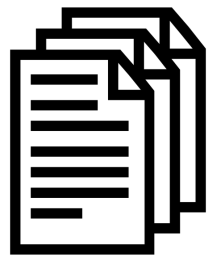
```
{'atom': 4.0140, 'bomb': 4.0704, 'bring': 2.7239, 'continu':  
2.4331, 'end': 2.1559, 'energi': 2.5045, 'have': 1.0742, 'help':  
1.8157, 'histori': 2.4213, 'ii': 3.0998, 'impact': 3.0304, 'it':  
2.0473, 'legaci': 4.1335, 'manhattan': 4.1345, 'peac': 3.5205,  
'project': 2.6442, 'scienc': 2.8700, 'us': 0.9967, 'war':  
2.6454, 'world': 1.9974}
```



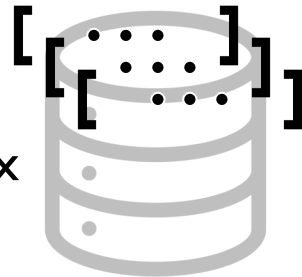
Results

“bag of words”
“lexical”
sparse vector

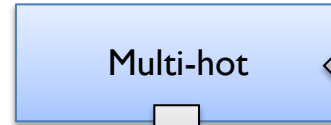
“Documents”



Term Weighting



Inverted Index



Query



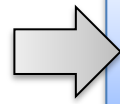
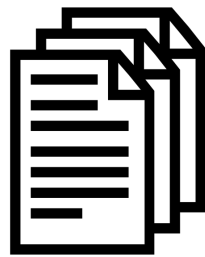
atomic bomb

{ 'atom': 1, 'bomb': 1 }

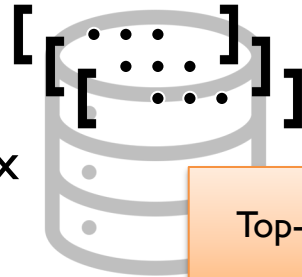


Results

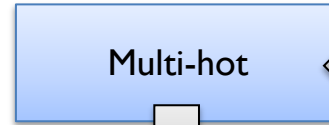
“Documents”



Term Weighting



Inverted Index



Multi-hot



[... q]

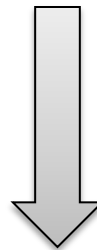
Top-k Retrieval

Query



atomic bomb

{ 'atom': 1, 'bomb': 1 }



Results

“Documents”



Term Weighting

Multi-hot

Query



atomic bomb

The Manhattan Project and its atomic bomb helped bring an end to World War II...

```
{'atom': 4.0140, 'bomb': 4.0704, 'bring': 2.7239, 'continu': 2.4331, 'end': 2.1559, 'energi': 2.5045, 'have': 1.0742, 'help': 1.8157, 'histori': 2.4213, 'ii': 3.0998, 'impact': 3.0304, 'it': 2.0473, 'legaci': 4.1335, 'manhattan': 4.1345... }
```

Top-k Retrieval

[... q]

```
{'atom': 1, 'bomb': 1}
```

inner (dot) product



Results

“Documents”



Term Weighting

Multi-hot

Query



atomic bomb

The Manhattan Project and its atomic bomb helped bring an end to World War II...

```
{ 'atom': 4.0140, 'bomb': 4.0704, 'bring': 2.7239, 'continu': 2.4331, 'end': 2.1559, 'energi': 2.5045, 'have': 1.0742, 'help': 1.8157, 'histori': 2.4213, 'ii': 3.0998, 'impact': 3.0304, 'it': 2.0473, 'legaci': 4.1335, 'manhattan': 4.1345... }
```

Top-k Retrieval

[... q]

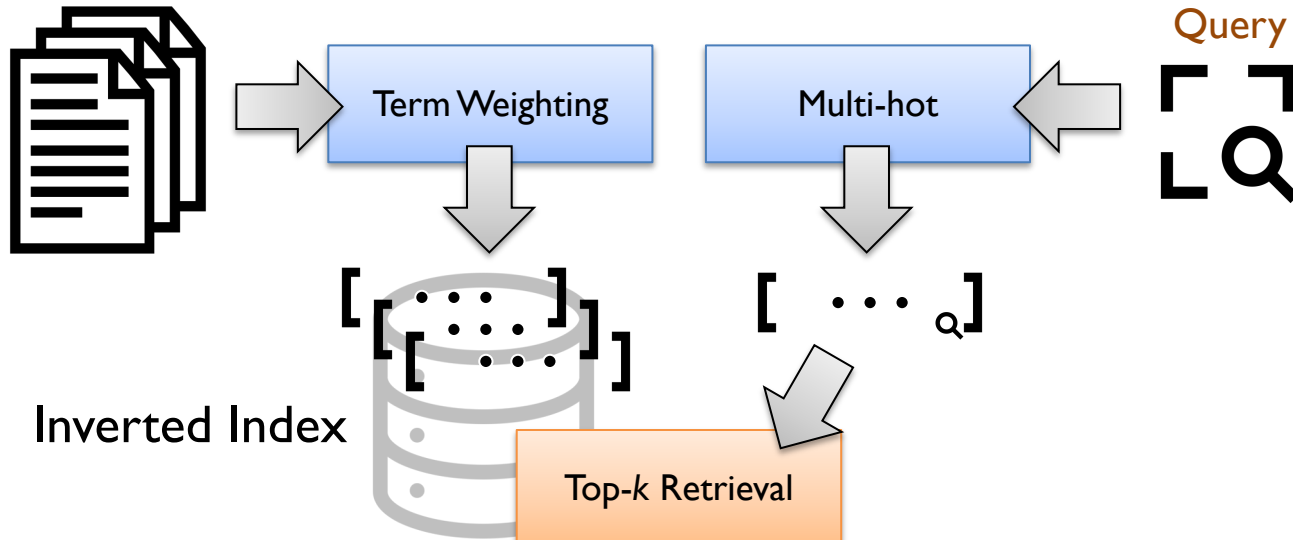
```
{ 'atom': 1, 'bomb': 1 }
```

inner (dot) product



Results

“Documents”



Inverted Index

tl;dr – retrieval ~ computing
dot products on vector
representations!



Results

“Documents”



Term Weighting

Multi-hot

Query



atomic bomb

The Manhattan Project and its atomic bomb helped bring an end to World War II...

```
{ 'atom': 4.0140, 'bomb': 4.0704, 'bring': 2.7239, 'continu': 2.4331, 'end': 2.1559, 'energi': 2.5045, 'have': 1.0742, 'help': 1.8157, 'histori': 2.4213, 'ii': 3.0998, 'impact': 3.0304, 'it': 2.0473, 'legaci': 4.1335, 'manhattan': 4.1345... }
```

Top-k Retrieval

[... q]

```
{ 'atom': 1, 'bomb': 1 }
```

inner (dot) product



Results

“Documents”



Term Weighting

Multi-hot

Query



nuclear weapon

The Manhattan Project and its atomic bomb helped bring an end to World War II...

```
{ 'atom': 4.0140, 'bomb': 4.0704, 'bring': 2.7239, 'continu': 2.4331, 'end': 2.1559, 'energi': 2.5045, 'have': 1.0742, 'help': 1.8157, 'histori': 2.4213, 'ii': 3.0998, 'impact': 3.0304, 'it': 2.0473, 'legaci': 4.1335, 'manhattan': 4.1345... }
```

Top-k Retrieval

```
{ 'nuclear': 1, 'weapon': 1 }
```

inner (dot) product

... which is?



Results

O

Semantic Mismatch

bag-of-words representations rely on lexical overlap

“Documents”



Term Weighting

Multi-hot

Query



nuclear weapon

The Manhattan Project and its atomic bomb helped bring an end to World War II...

```
{ 'atom': 4.0140, 'bomb': 4.0704, 'bring': 2.7239, 'continu': 2.4331, 'end': 2.1559, 'energi': 2.5045, 'have': 1.0742, 'help': 1.8157, 'histori': 2.4213, 'ii': 3.0998, 'impact': 3.0304, 'it': 2.0473, 'legaci': 4.1335, 'manhattan': 4.1345... }
```

Top-k Retrieval

```
{ 'nuclear': 1, 'weapon': 1 }
```

inner (dot) product = 0



Results

But...

“Documents”



Term Weighting

Multi-hot

Query



nuclear weapon

The Manhattan Project and its atomic bomb helped bring an end to World War II...

“meaning”

```
{ 'atom': 4.0140, 'bomb': 4.0704, 'bring': 2.7239, 'continu': 2.4331, 'end': 2.1559, 'energi': 2.5045, 'have': 1.0742, 'help': 1.8157, 'histori': 2.4213, 'ii': 3.0998, 'impact': 3.0304, 'it': 2.0473, 'legaci': 4.1335, 'manhattan': 4.1345... }
```

Top-k Retrieval

“meaning”
{ 'nuclear': 1, 'weapon': 1 }

Semantic matching?



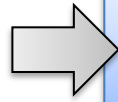
Results

What if representations can capture “meaning”?

What type of representations?

What does “meaning” mean?

“Documents”

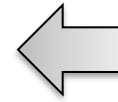


Term Weighting



[...]

Query



The Manhattan Project and its atomic bomb helped bring an end to World War II. Its legacy of peaceful uses of atomic energy continues to have an impact on history and science.

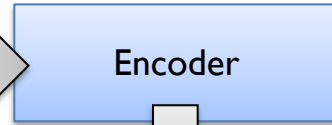
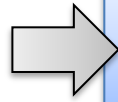
```
{'atom': 4.0140, 'bomb': 4.0704, 'bring': 2.7239, 'continu':  
2.4331, 'end': 2.1559, 'energi': 2.5045, 'have': 1.0742, 'help':  
1.8157, 'histori': 2.4213, 'ii': 3.0998, 'impact': 3.0304, 'it':  
2.0473, 'legaci': 4.1335, 'manhattan': 4.1345, 'peac': 3.5205,  
'project': 2.6442, 'scienc': 2.8700, 'us': 0.9967, 'war':  
2.6454, 'world': 1.9974}
```

sparse vector



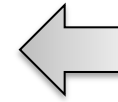
Results

“Documents”



[...]

Query



The Manhattan Project and its atomic bomb helped bring an end to World War II. Its legacy of peaceful uses of atomic energy continues to have an impact on history and science.

```
[0.099843978881836, 0.8700575828552246, 0.520509719848633,  
0.030491352081299, 0.7239298820495605, 0.134523391723633,  
0.4331274032592773, 0.644286632537842, 0.645430564880371,  
0.0473427772521973, 0.070496082305908, 0.504533529281616,  
0.8157329559326172, 0.133575916290283, 0.9974448680877686,  
0.0742542743682861, 0.1559412479400635, 0.421395778656006,  
0.014032363891602, 0.996794581413269...]
```



Results

dense vector

What if representations can capture “meaning”?

What type of representations?

What does “meaning” mean?

What if...

vectors of queries and relevant passages get high inner products
vectors of queries and non-relevant passages get low inner products

And you have lots of examples?

This sounds an awful lot like...

A machine learning problem!



What if representations can capture “meaning”?

What type of representations?

What does “meaning” mean?

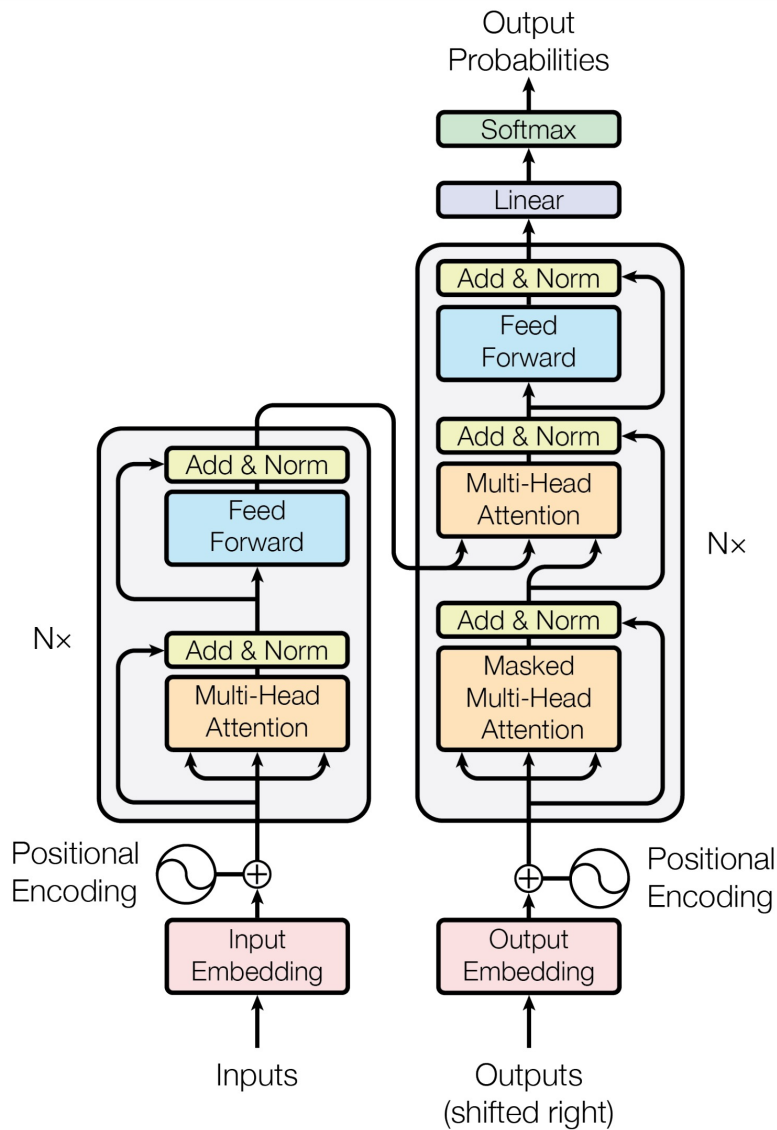
What if...

vectors of queries and relevant passages get high inner products
vectors of queries and non-relevant passages get low inner products

And you have lots of examples?

learned representations
= *embeddings*

But how?



Transformers!

Attention Is All You Need

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Jakob Uszkoreit*
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Llion Jones*
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Aidan N. Gomez*[†]
University of Toronto
aidan@cs.toronto.edu

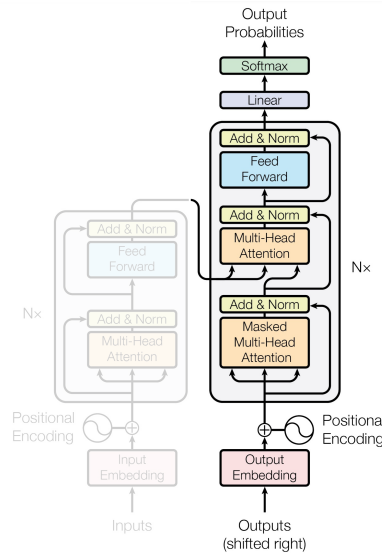
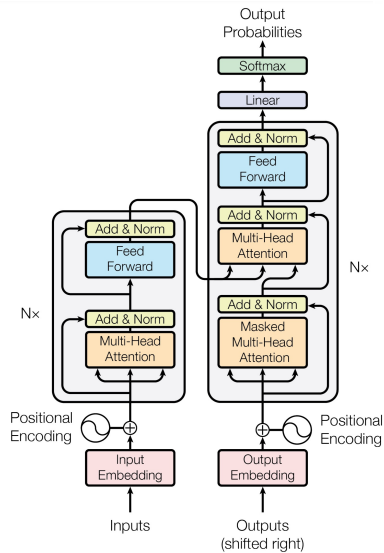
Łukasz Kaiser*
Google Brain
lukaszkaiser@google.com

Illia Polosukhin*[‡]
illia.polosukhin@gmail.com

Abstract

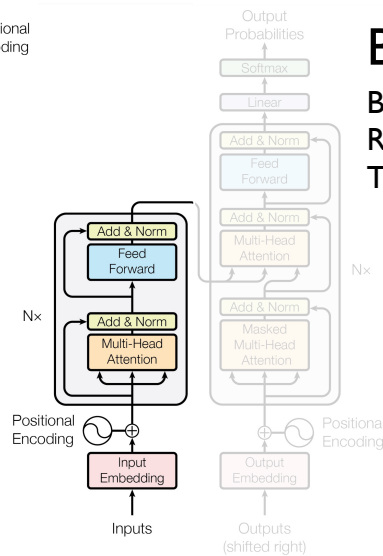
The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task,

Transformer (2017)



GPT (2018)

Generative Pretrained Transformer

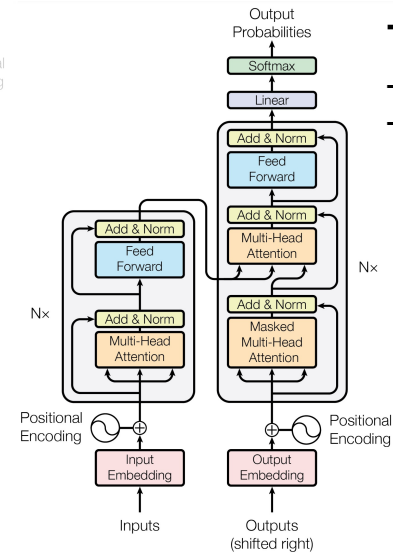


BERT (2018)

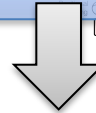
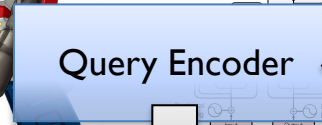
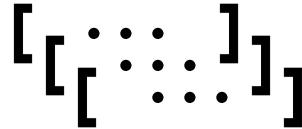
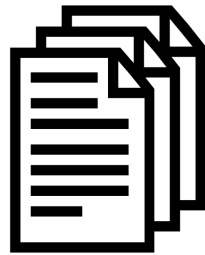
Bidirectional Encoder Representations from Transformers

T5 (2019)

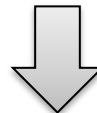
Text-To-Text Transfer Transformer



“Documents”

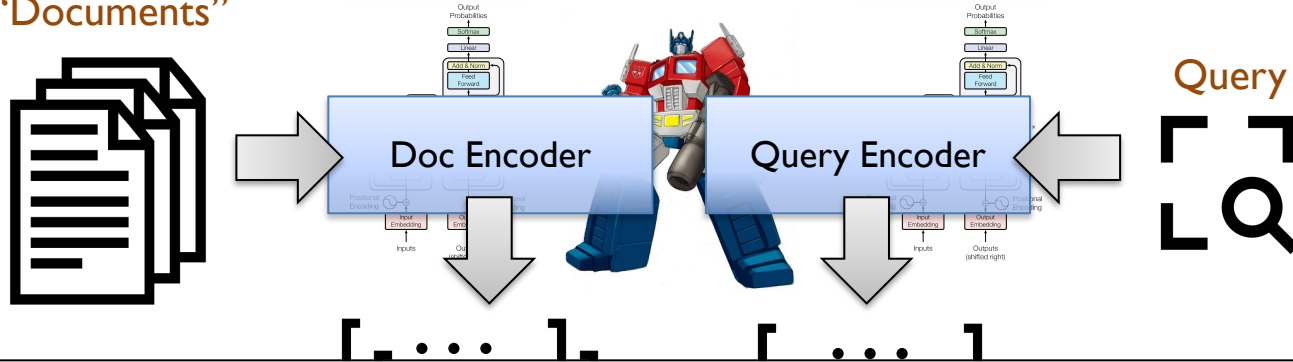


Query



Results

“Documents”



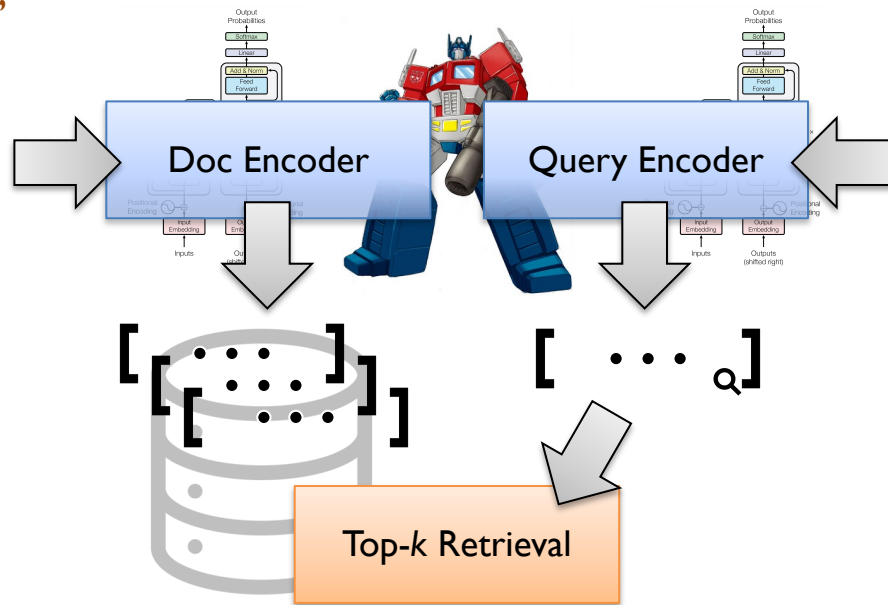
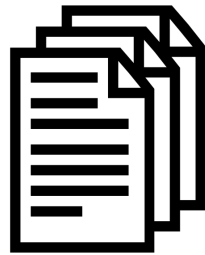
The Manhattan Project and its atomic bomb helped bring an end to World War II. Its legacy of peaceful uses of atomic energy continues to have an impact on history and science.

```
[0.099843978881836, 0.8700575828552246, 0.520509719848633,  
0.030491352081299, 0.7239298820495605, 0.134523391723633,  
0.4331274032592773, 0.644286632537842, 0.645430564880371,  
0.0473427772521973, 0.070496082305908, 0.504533529281616,  
0.8157329559326172, 0.133575916290283, 0.9974448680877686,  
0.0742542743682861, 0.1559412479400635, 0.421395778656006,  
0.014032363891602, 0.996794581413269...]
```



Results

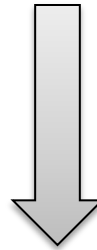
“Documents”



Query



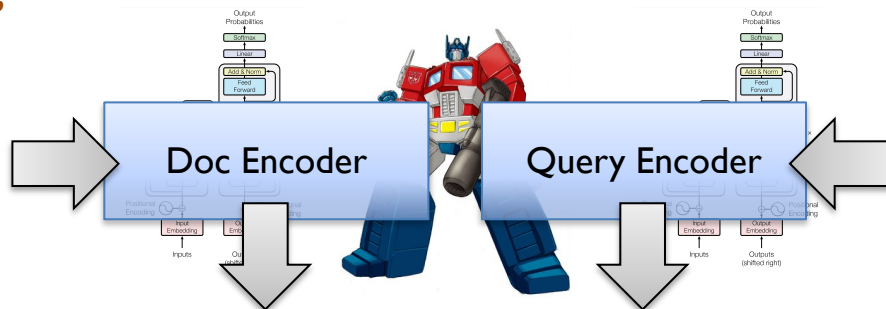
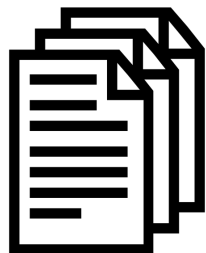
This is vector search!



Results

But wait, there's more!

“Documents”



Query



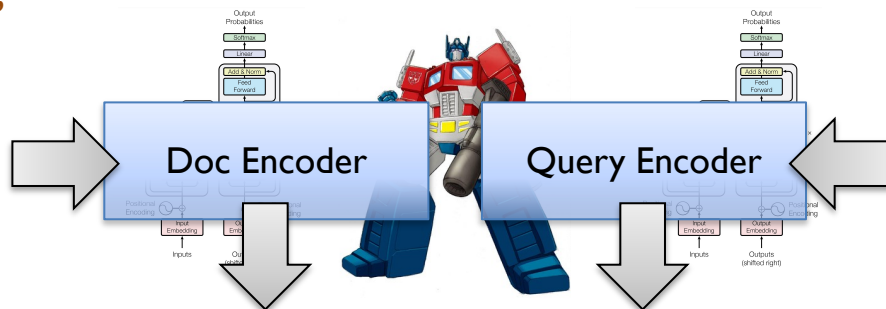
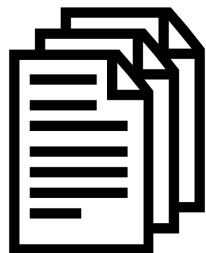
Top-k Retrieval

Reranking



Results

“Documents”



Query

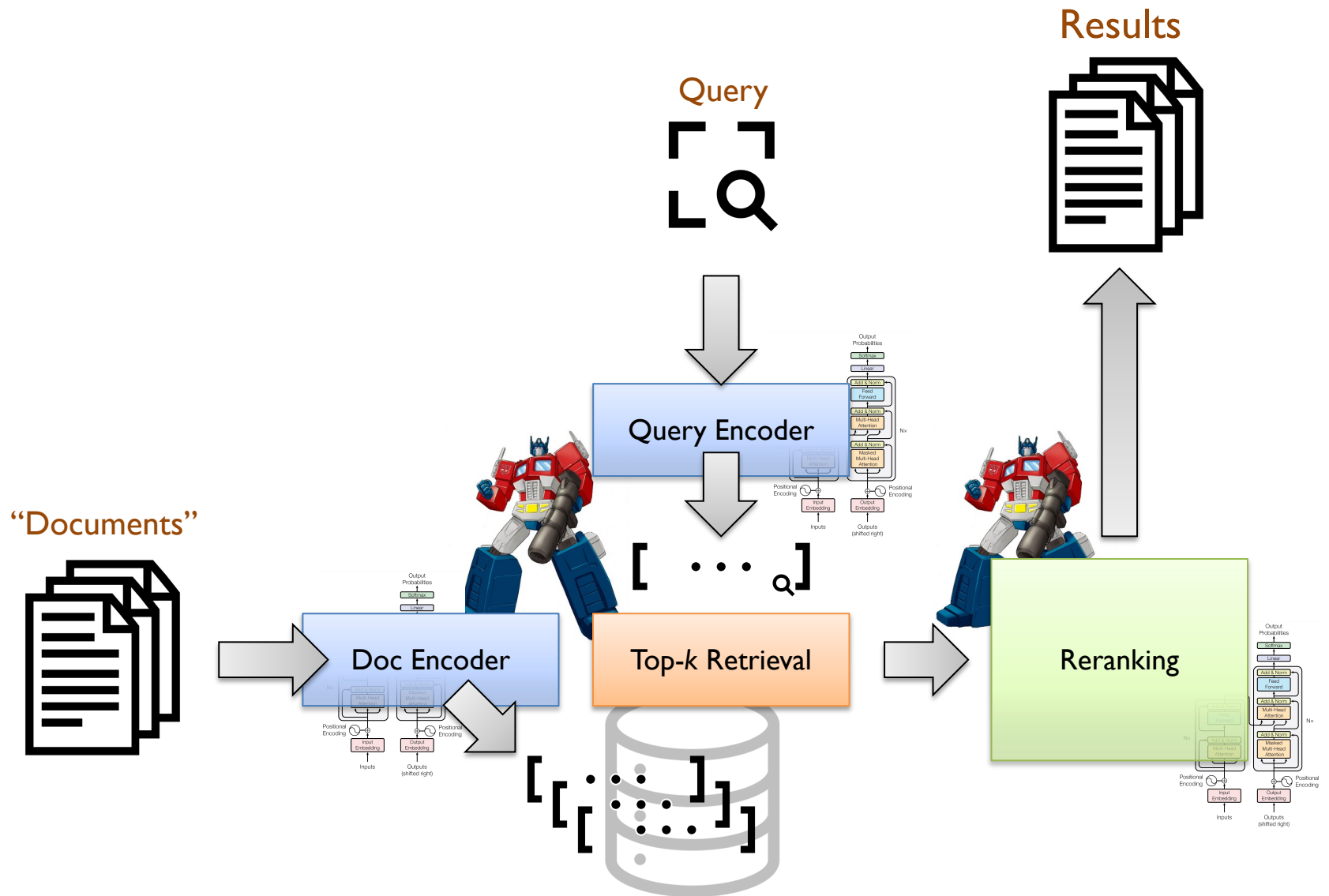


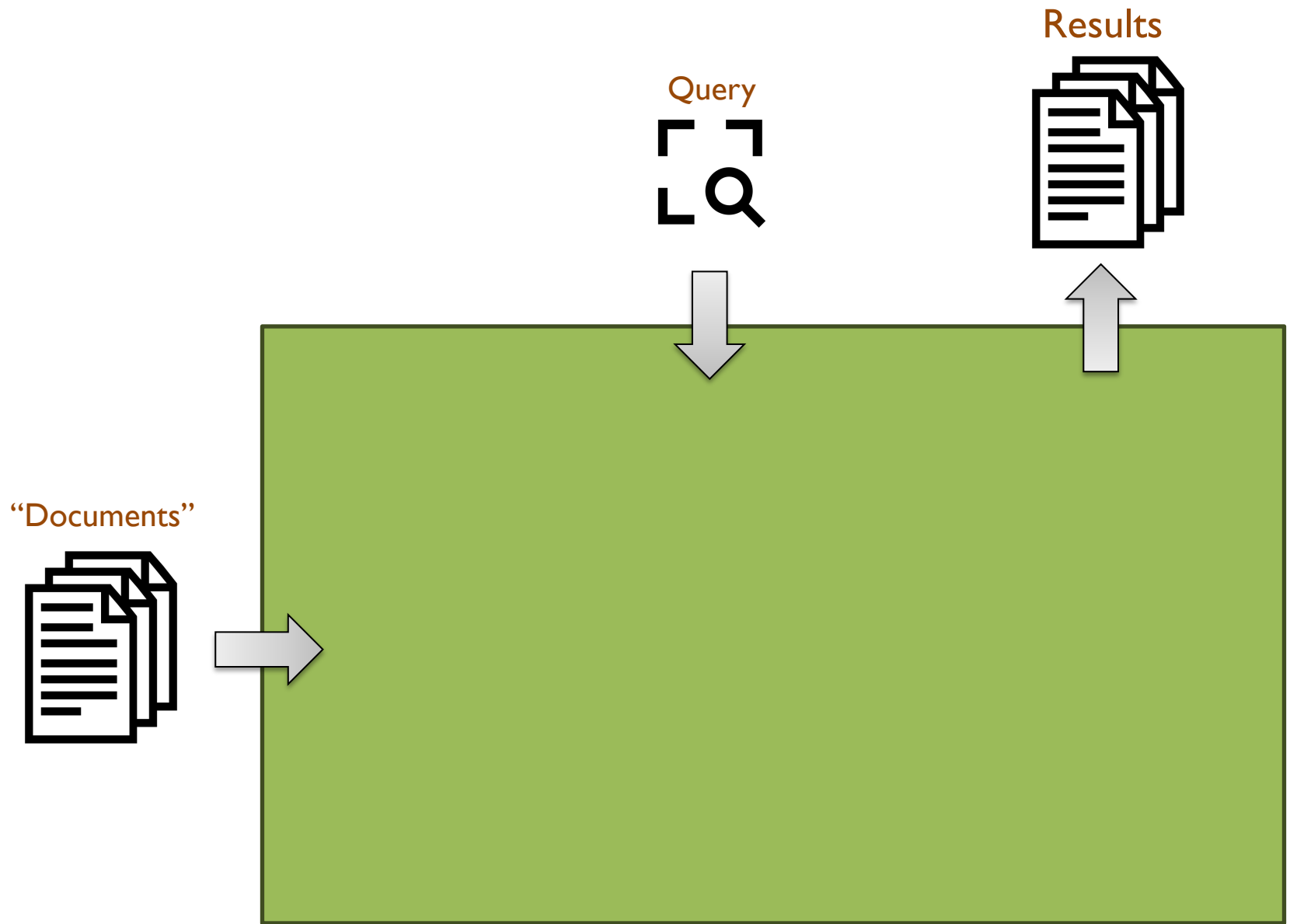
Top-k Retrieval

Reranking

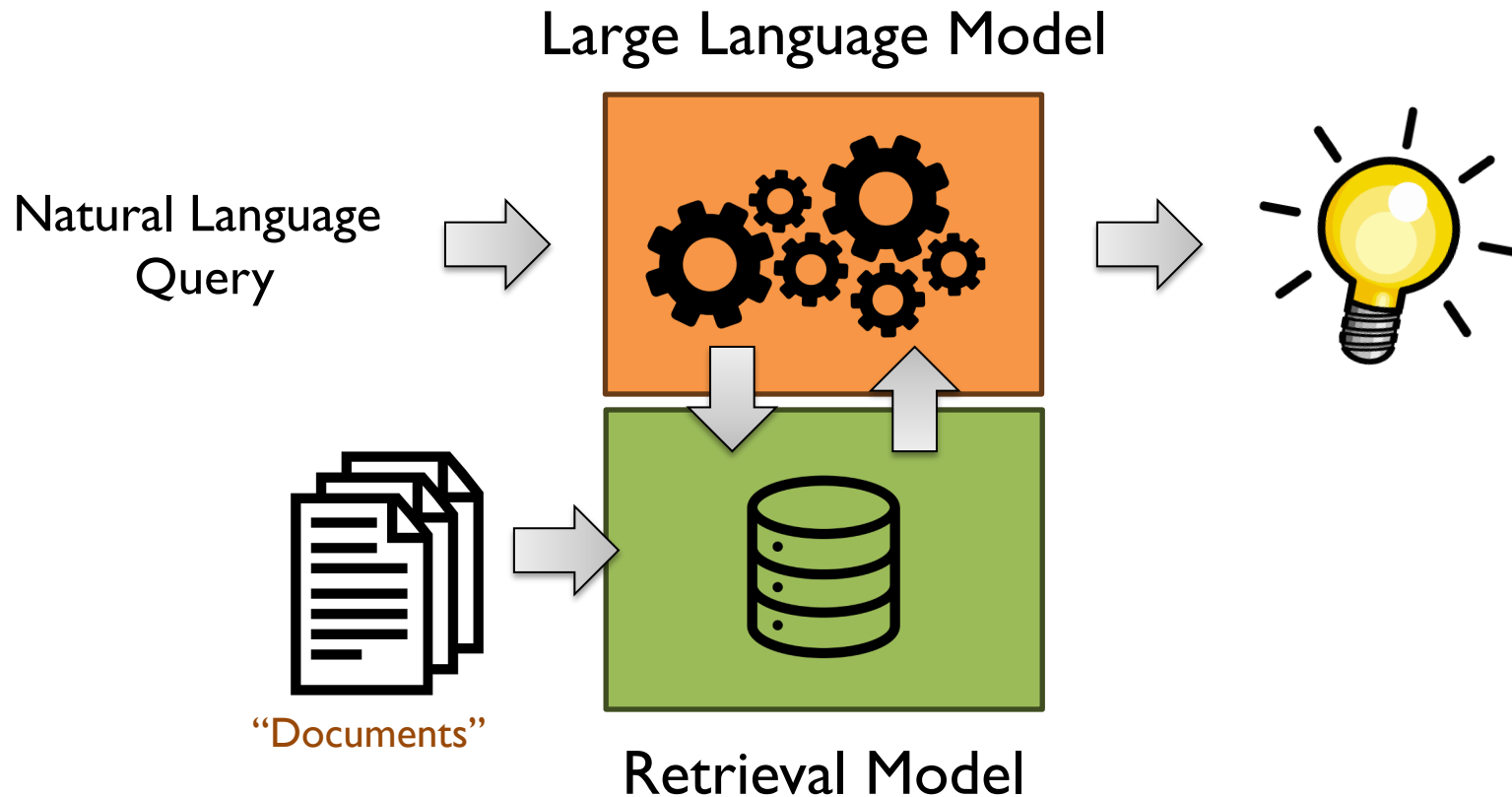


Results



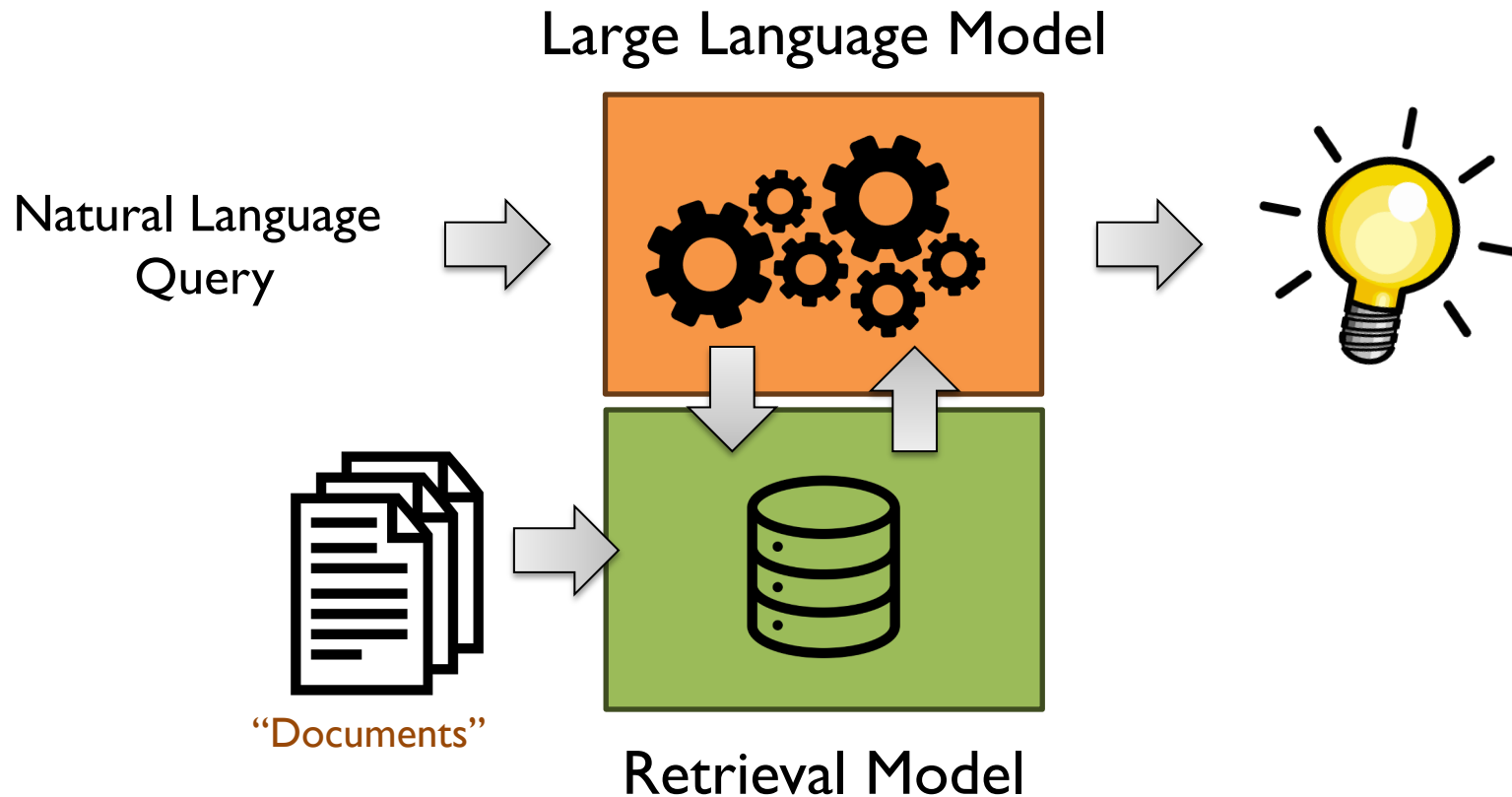


The Big Picture

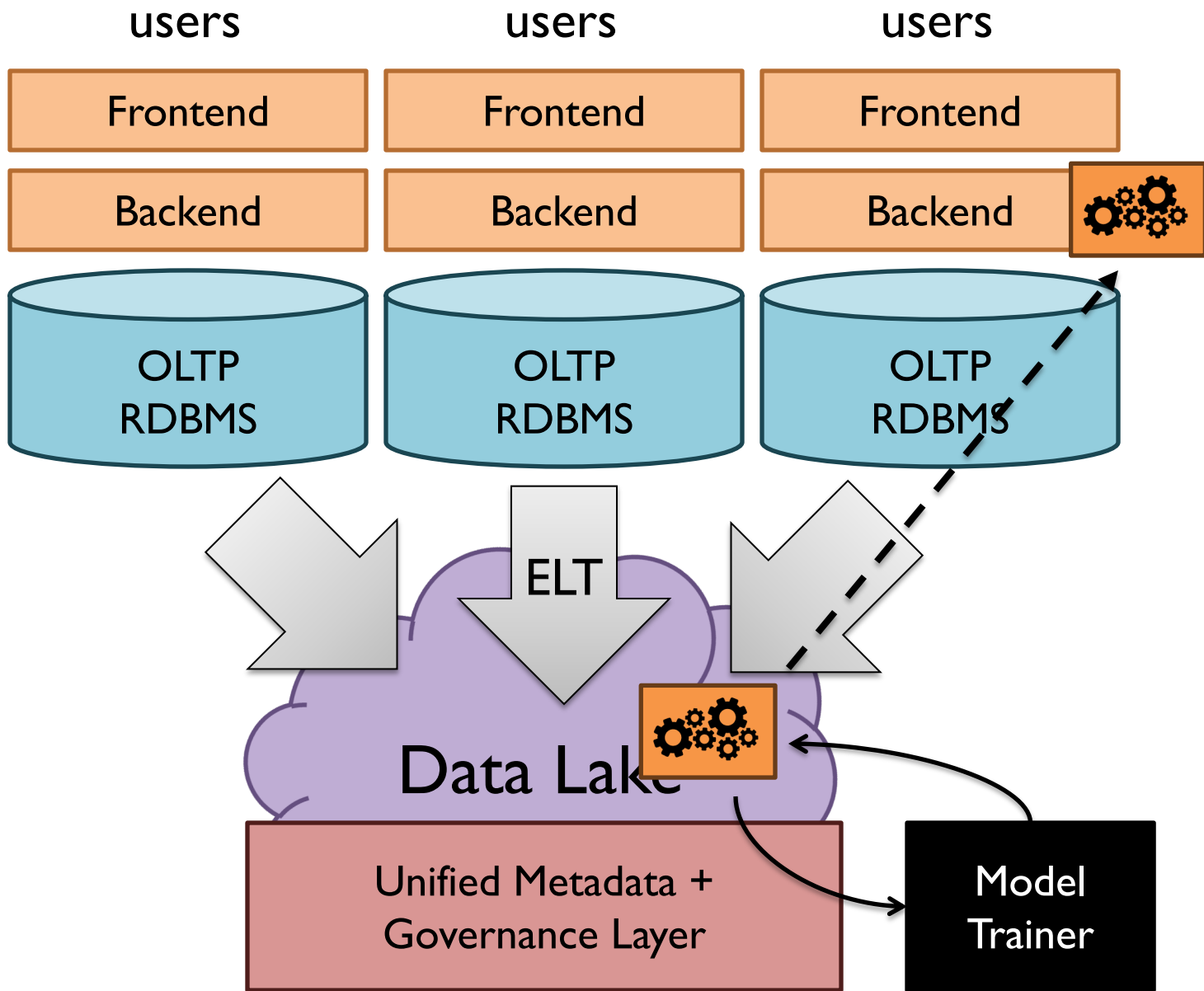


The Big Picture **Why?**

(combat) Hallucinations
(incorporate) Up-to-date information
(exploit) Private data



RAG / Lakehouse Integration: Why?



Lakehouse

RAG / Lakehouse Integration: Why?

Ingesting multiple sources of data

User-generated content + behavioral data

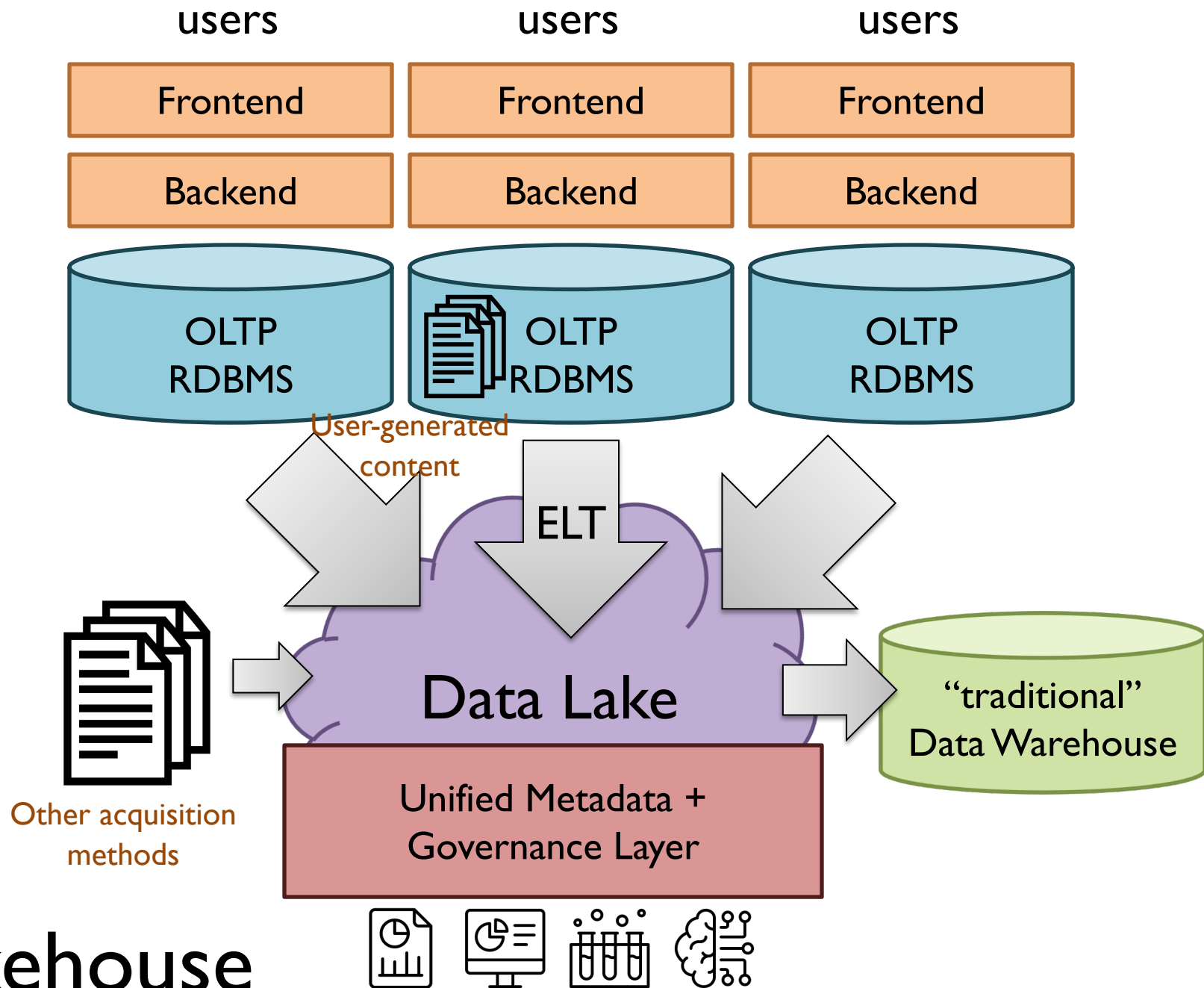
Other sources...

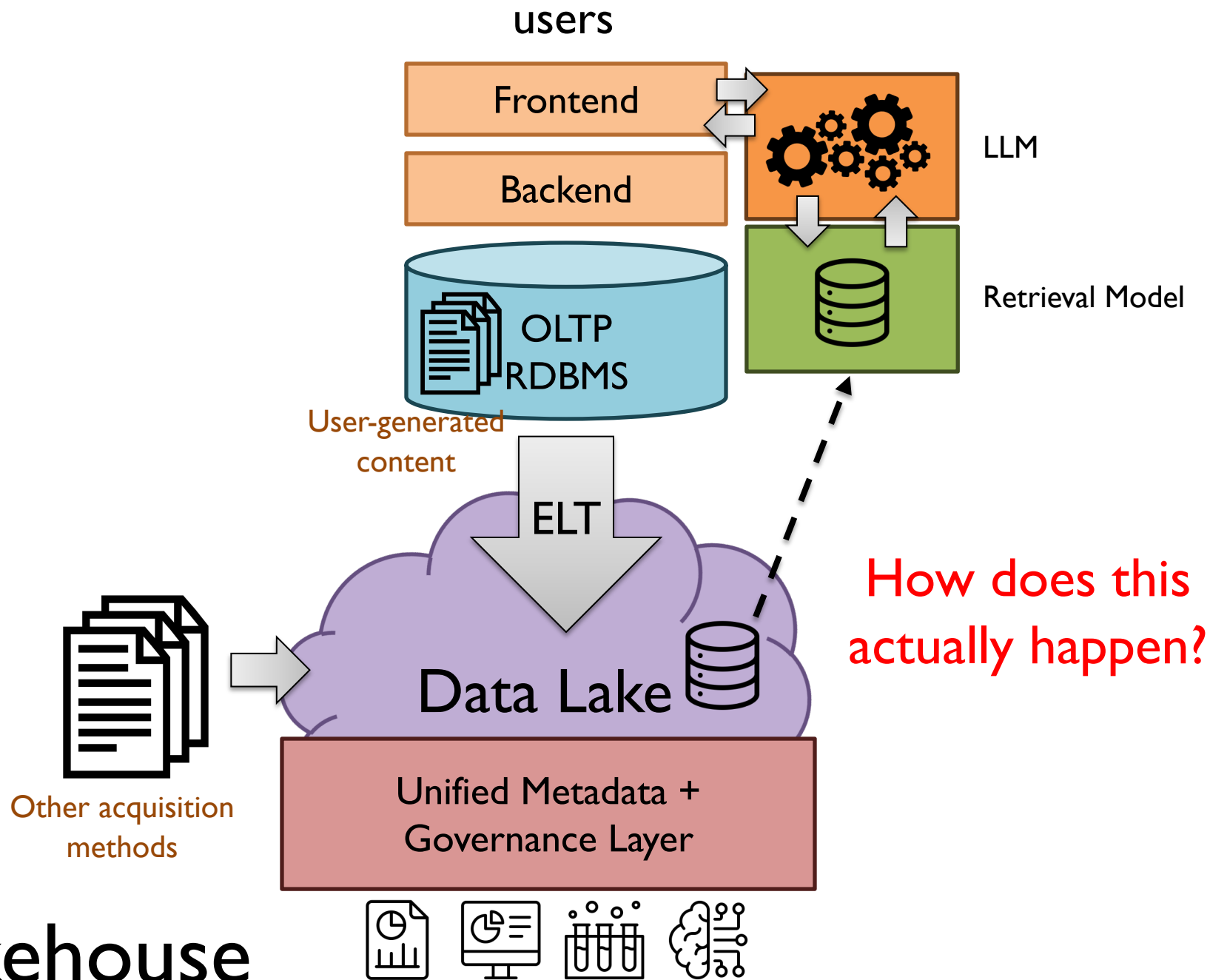
Performing “standard” lakehouse tasks

Joining, filtering, projecting, etc. heterogenous data

Data cleaning, aggregation, etc.

Training ML models





富嶽三十六景 神奈川
浪裏

